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A versatile ray-tracing code for studying rf wave propagation in toroidal magnetized plasmas

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Abstract
A new ray-tracing code named $C3PO$ has been developed to study the propagation of arbitrary electromagnetic radio-frequency (rf) waves in magnetized toroidal plasmas. Its structure is designed for maximum flexibility regarding the choice of coordinate system and dielectric model. The versatility of this code makes it particularly suitable for integrated modeling systems. Using a coordinate system that reflects the nested structure of magnetic flux surfaces in tokamaks, fast and accurate calculations inside the plasma separatrix can be performed using analytical derivatives of a spline-Fourier interpolation of the axisymmetric toroidal MHD equilibrium. Applications to reverse field pinch magnetic configuration are also included. The effects of 3D perturbations of the axisymmetric toroidal MHD equilibrium, due to the discreteness of the magnetic coil system or plasma fluctuations in an original quasi-optical approach, are also studied. Using a Runge–Kutta–Fehlberg method for solving the set of ordinary differential equations, the ray-tracing code is extensively benchmarked against analytical models and other codes for lower hybrid and electron cyclotron waves.

(Some figures may appear in colour only in the online journal)

1. Introduction

Radio-frequency (rf) waves are routinely used to heat and drive a current in toroidal plasmas. Describing the propagation of low-frequency waves (ion cyclotron wave, fast wave), for which the wavelength λ and the equilibrium variation scale length $\mathcal{L}$ are of the same order, generally requires full-wave techniques. For high-frequency rf waves—lower-hybrid (LH), electron cyclotron (EC) or electron Bernstein waves (EBW)—with $\lambda \ll \mathcal{L}$, ray-tracing techniques, which typically require much less computing time and resources, are adequate if the Wentzel–Kramers–Brillouin (WKB) approximation is valid [1–5]. Under these conditions, solving Maxwell equations becomes a standard initial value problem with a set of coupled ordinary differential equations.

Many ray-tracing codes in axisymmetric toroidal plasmas use the toroidal coordinates system (R, Z, ϕ), which is attractive because its orthogonality results in a simplified implementation of the differential equations [5–7]. However, since these coordinates are not specific to the topology of the plasma inside the separatrix, characterized by nested magnetic flux surfaces, repetitive and time-consuming two-dimensional local interpolations must be performed, with limited accuracy concerning numerical derivatives. The corresponding cumulative numerical error can result in a significant deviation of the ray trajectory before full absorption of the rf wave power.

Under the framework of integrated modeling where ray-tracing calculations can be part of a complex chain of codes, computing time and robustness become essential in the design of a numerical tool [8]. Taking into consideration these constraints, the use of magnetic flux coordinates (ψ, θ, ϕ) is an important advantage despite the more complicated metrics. First, only 1D interpolations are required for flux functions, such as the equilibrium density and temperature. Secondly, the poloidal direction described by the angular coordinate θ uses Fourier series for fast and accurate interpolations and the calculation of derivatives to any order. The description of the

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toroidal MHD equilibrium is based on a non-local interpolation of its discrete form by cubic spline in the radial direction and Fourier series in the poloidal direction. It guarantees high accuracy and global smoothness of all interpolated quantities and their derivatives and is well suited for ray-tracing calculations. In addition, the poloidal periodicity is naturally satisfied in this approach, an important criterion when rays undergo several poloidal circumvolutions before the power carried by the rf wave is fully damped. Moreover, this system of coordinates is also used by a variety of codes such as toroidal MHD equilibrium codes and Fokker–Planck solvers [9, 10]. Coupling the ray-tracing solver to such codes may be done without any time-consuming coordinate conversions and grid resampling, which can also introduce substantial additional numerical errors.

The new ray-tracing code presented here, which is a module of the CRONOS tokamak simulation code [11], is designed for maximum flexibility regarding the shape of the plasma cross-section and the dielectric model. Based on the curvilinear coordinate system (ψ, θ, ϕ) associated with a axisymmetric toroidal MHD equilibrium with nested magnetic flux surfaces, this code is primarily dedicated—though not restricted—to study propagation of any rf wave in the volume circumvented by the magnetic separatrix, where most of the power is expected to be absorbed [12]. An important feature of this code is that 3D equilibrium perturbations can be included while taking advantage of the high efficiency of its numerical scheme. The rf wave propagation can thus be calculated in the presence of drift wave-type fluctuations or the toroidal magnetic field ripple due to the discrete number of coils [13].

In addition to ray trajectories, the code calculates wave properties of interest for heating and current drive calculations such as the polarization, energy flow and linear power absorption [1]. Based on a Runge–Kutta–Fehlberg method for solving the set of ordinary differential equations [14, 15], this ray-tracing code named *C3PO* is written in *C* language and also distributed as a mexfile for including in Matlab®[3]. It is benchmarked extensively for various types of waves against both existing codes and analytical models.

In section 2, the basics of the ray-tracing theory are briefly revisited, with particular emphasis on the possibility to build a modular code based on full separation between the metrics and the physics in the plasma. Then, the curvilinear representation associated with the toroidal MHD equilibrium is introduced in section 3. The initial and boundary conditions problems are discussed in section 4. After introducing the numerical methods in section 5, the verification of *C3PO* is obtained through benchmarking against analytical models and other ray-tracing codes in section 6. Finally, some illustrative examples highlighting the capabilities of the code *C3PO* are shown in section 7 before the conclusion.

2. Ray tracing in a slowly varying plasma

From linear wave theory in an infinite uniform plasma, the Fourier transform of the wave electric field $\boldsymbol{E}_{\boldsymbol{k}}$ is the solution of the wave equation $\mathbb{D} \cdot \boldsymbol{E}_{\boldsymbol{k}} = 0$, where $\mathbb{D} = \boldsymbol{N}\boldsymbol{N} - N^2\mathbb{I} + \mathbb{K}(\boldsymbol{N}, \omega)$ is the Maxwell or dispersion tensor. In $\mathbb{D}$, $\boldsymbol{N} \equiv c\boldsymbol{k}/\omega$ is the index of refraction, where $\boldsymbol{k}$ and ω are the wave vector and frequency of the excited rf wave, respectively. The permittivity tensor is given by the relation $\mathbb{K}(\boldsymbol{N}, \omega) = \mathbb{I} + \mathbb{X}(\boldsymbol{N}, \omega)$.

The susceptibility tensor $\mathbb{X}$ describes the linear response of the plasma to the rf wave perturbation in an equilibrium magnetic field $\boldsymbol{B} = B\hat{\boldsymbol{b}}$ with $B = \|\boldsymbol{B}\|$. It is the sum of contributions from all species $\mathbb{X} = \sum_s \mathbb{X}^s$. In the case where the distribution function of the species s (with mass m_s and charge q_s) is a Maxwellian distribution characterized by a density n_s and a temperature T_s, $\mathbb{X}^s$ depends upon the index of refraction $\boldsymbol{N}$, the thermal velocity $\beta_{T_s} = v_{T_s}/c$ normalized to the speed of light c, where $v_{T_s} = \sqrt{kT_s/m_s}$, and the normalized frequencies $\hat{\omega}_{\mathrm{p}s} = \omega_{\mathrm{p}s}/\omega$ and $\hat{\omega}_{\mathrm{c}s} = \omega_{\mathrm{c}s}/\omega$ which are all dimensionless parameters. Here, $\omega_{\mathrm{p}s} = \sqrt{q_s^2 n_s/\varepsilon_0 m_s}$ is the plasma frequency, $\omega_{\mathrm{c}s} = q_s B/m_s$ is the cyclotron frequency. In vacuum, $\mathbb{K} = \mathbb{I}$ since $\mathbb{X} = 0$.

In order to obtain non-trivial solutions to the wave equation, the determinant of $\mathbb{D}$ must be zero, $\mathcal{D}(\boldsymbol{N}, \beta_{Ts}, \hat{\omega}_{\mathrm{p}s}, \hat{\omega}_{\mathrm{c}s}) \equiv \det \mathbb{D}(\boldsymbol{N}, \beta_{Ts}, \hat{\omega}_{\mathrm{p}s}, \hat{\omega}_{\mathrm{c}s}) = 0$, which defines the dispersion relation. It provides the local electromagnetic eigenmodes that can be excited independently in the homogeneous plasma [17]. By cylindrical symmetry about the local direction of the magnetic field $\hat{\boldsymbol{b}}$, $\mathcal{D}$ only depends upon the components of $\boldsymbol{N}$ parallel and perpendicular to the magnetic field, defined, respectively, as $N_\parallel = \boldsymbol{N} \cdot \hat{\boldsymbol{b}}$ and $N_\perp = \|\boldsymbol{N} \times \hat{\boldsymbol{b}}\|$. Thus, the dispersion relation can be solved for either $\omega(N_\parallel, N_\perp)$, $N_\parallel(\omega, N_\perp)$ or $N_\perp(\omega, N_\parallel)$. For benchmarking purpose, only the vacuum and cold plasma model are considered here, even though kinetic plasma models are implemented in the code. Finally, the polarization of eigenmodes is determined from the corresponding eigenvectors of the wave equation.

Assuming that ω and $N_\parallel$ are given real quantities, the condition $\mathcal{D} = 0$ generally leads to a complex $N_\perp = N_{\perp r} + \mathrm{i}N_{\perp i}$. Meanwhile, the dispersion tensor $\mathbb{D}$ can be generally decomposed as $\mathbb{D} = \mathbb{D}^{\mathrm{H}} + \mathrm{i}\mathbb{D}^{\mathrm{A}}$ where $\mathbb{D}^{\mathrm{H}} = (\mathbb{D} + \mathbb{D}^\dagger)/2$ and $\mathbb{D}^{\mathrm{A}} = -\mathrm{i}(\mathbb{D} - \mathbb{D}^\dagger)/2$ are the Hermitian and anti-Hermitian parts of $\mathbb{D}$, respectively. In this work, the weak damping approximation is considered[4], which is valid if $|\, D_{ij}^{\mathrm{A}} \,| \ll |\, D_{ij}^{\mathrm{H}} \,|$ [18]. In that case, it can be shown that $|\, N_{\perp i} \,| \ll |\, N_{\perp r} \,|$ and $N_{\perp r}$ can be determined by solving the approximate wave equation $\mathbb{D}^{\mathrm{H}}(N_{\perp r}) \cdot \boldsymbol{E}_{\boldsymbol{k}} = 0$ with the corresponding dispersion relation $\mathcal{D}^{\mathrm{H}} \equiv \det \mathbb{D}^{\mathrm{H}} = 0$. In this approximation, the time-averaged density of the energy flow is $\boldsymbol{S} = -(\varepsilon_0 c/4)\partial(\boldsymbol{E}_{\boldsymbol{k}}^* \cdot \mathbb{D}^{\mathrm{H}} \cdot \boldsymbol{E}_{\boldsymbol{k}})/\partial \boldsymbol{N}$ and the time-averaged density of dissipated power P can be obtained from the relation $P = (\varepsilon_0 \omega/2)\boldsymbol{E}_{\boldsymbol{k}}^* \cdot \mathbb{D}^{\mathrm{A}} \cdot \boldsymbol{E}_{\boldsymbol{k}}$. The imaginary part of the wave vector, which yields the ray damping, is calculated using $N_{\perp i} = (c/2\omega)P/S_\perp$ with $S_\perp = \|\boldsymbol{S} \times \hat{\boldsymbol{b}}\|$. From here on in this paper, $\boldsymbol{k}$, $\boldsymbol{N}$ and $\boldsymbol{E}_{\boldsymbol{k}}$ will refer to the zero-order solution of $\mathbb{D}^{\mathrm{H}}(N_{\perp r}) \cdot \boldsymbol{E}_{\boldsymbol{k}} = 0$ with

[3] Release R2007b or later [16].

[4] In the very near vicinity of the EC resonance layer, the weak damping approximation breaks down. However, this limit is never effectively reached, even for the EC wave, since the absorption width Δ always exceeds the EC wavelength λ, even in the linear limit. The breakdown of the weak damping approximation is critical only if $\Delta \sim \lambda$, which is so far never the case in realistic modeling of EC heating and current drive in tokamak. In the very near vicinity of the resonance (at the length scale λ), the power carried by the wave is already fully transferred to the particles.

respect to the small parameter $\epsilon \sim \max(|\mathbb{D}_{ij}^A|/|\mathbb{D}_{ij}^H|) \ll 1$ and the dispersion relation $\mathcal{D}$ to the Hermitian part $\mathcal{D}^{\mathrm{H}}$. The weak damping approximation separates the propagation and the absorption processes in rf wave calculations [19].

The ray-tracing representation of wave propagation is based on the validity of the WKB asymptotic expansion such that wave characteristics that usually apply to uniform plasmas (Fourier space description, group velocity, conductivity tensor) can be considered locally in a slowly inhomogeneous plasma. The spatial ordering $\lambda \ll \mathcal{L}$ is a necessary condition to be satisfied, where λ is the wavelength and $\mathcal{L}$ is the shortest local gradient length in the plasma, i.e. $\mathcal{L}^{-1} \equiv \min(\|\boldsymbol{\nabla}\beta_{Ts}\|/\beta_{Ts}, \|\boldsymbol{\nabla}\hat{\omega}_{\mathrm{ps}}\|/\hat{\omega}_{\mathrm{ps}}, \|\boldsymbol{\nabla}\hat{\omega}_{\mathrm{cs}}\|/|\hat{\omega}_{\mathrm{cs}}|)$. The electric field can then be expressed as a wave packet $\boldsymbol{E}_{\mathrm{rf}}(\boldsymbol{x}, t) \equiv \mathcal{E}(\boldsymbol{x})\mathrm{e}^{\mathrm{i}S(\boldsymbol{x},\boldsymbol{k},\omega,t)}$ where $\mathcal{S}(\boldsymbol{x}, \boldsymbol{k}, \omega, t)$ is the phase or eikonal function, the wave vector $\boldsymbol{k} \equiv \nabla\mathcal{S}$ being much larger than the gradient of $\mathcal{E}$[5]. By introducing the wave vector as the gradient of the phase, it is assumed that the surface of constant phase is a smooth function in space, which introduces an additional condition on the applicability of the WKB approximation. Consequently, the ray tracing has lost its physical meaning if the wave absorption takes place after an integration time exceeding the maximal Liapunov exponent, which characterizes the rate of separation of infinitesimally close trajectories. Such a problem is particularly critical for the prediction of the current driven by LH wave in tokamaks, where the stochastic nature of the ray dynamics can limit the validity of the ray-tracing formalism [21–24].

Ray equations in a plasma slowly varying in space and time are defined by the condition $\delta\mathcal{S} = 0$ subject to $\mathcal{D}(\boldsymbol{X}, t, \boldsymbol{k}, \omega) = 0$. Since $\mathcal{S}$ is independent of the choice of the path in phase space, the coordinates k_j of the wave vector $\boldsymbol{k}$ must be canonically conjugate to those X_i of the position vector $\boldsymbol{X}$, i.e. $\{X_i, k_j\} = \delta_{ij}$, where $\{\ldots\}$ is the Poisson bracket and δ_{ij} is the Kronecker delta symbol. Similarly, the frequency ω with the time t are canonical conjugates. According to the Hamiltonian theory, the ray equations may be expressed in the general form

$$\frac{\mathrm{d}\boldsymbol{X}}{\mathrm{d}t} = -\frac{\partial\mathcal{D}}{\partial\boldsymbol{k}}\Big/\frac{\partial\mathcal{D}}{\partial\omega} \tag{1}$$

$$\frac{\mathrm{d}\boldsymbol{k}}{\mathrm{d}t} = \frac{\partial\mathcal{D}}{\partial\boldsymbol{X}}\Big/\frac{\partial\mathcal{D}}{\partial\omega} \tag{2}$$

found in the literature [4, 5, 21, 25]. The time evolution of the toroidal MHD equilibrium is considered to be much longer than the time scales associated with the wave propagation—the wave period and the wavepacket duration. In this case, typically verified for high-frequency rf waves (including in a fluctuating plasma), the time t may be considered as the evolution parameter as $\partial\mathcal{D}/\partial t = 0$, and ω is a constant of the ray dynamics imposed by the rf wave generator.

A 8-dimension pseudo-vector $\boldsymbol{Y}$ representing the phase-space coordinates is introduced

$$\boldsymbol{Y} = \begin{pmatrix} \boldsymbol{X} \\ \boldsymbol{k} \\ t \\ \omega \end{pmatrix} \tag{3}$$

[5] The identification of $\nabla\mathcal{S}$ with the wavevector $\boldsymbol{k}$ is possible if the condition $\nabla \cdot \boldsymbol{k} \ll \|\boldsymbol{k}\|^2$ is also fulfilled [20].

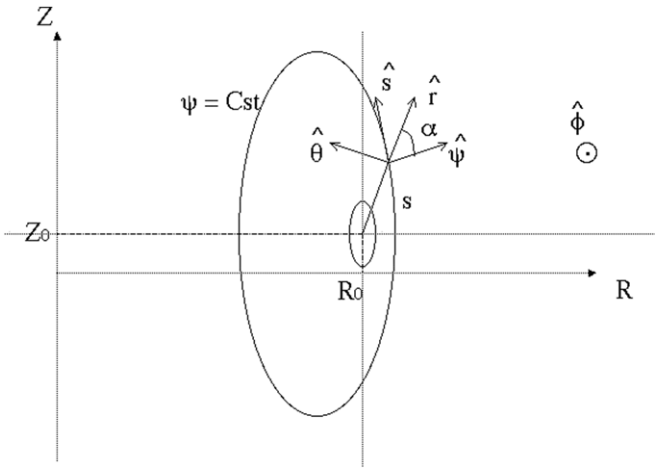


Figure 1. Magnetic flux coordinate system for $C3PO$ ray tracing.

and solving the dispersion relation requires explicitly the derivatives $\partial N_{\parallel}/\partial\boldsymbol{Y}$, $\partial N_{\perp}/\partial\boldsymbol{Y}$ and $\partial X_{ij}^s/\partial\boldsymbol{Y}$ with respect to the coordinates vector $\boldsymbol{Y}$.

Since the susceptibility tensor is only a function of the dimensionless parameters $(N_{\perp}, N_{\parallel}, \beta_{Ts}, \bar{\omega}_{\mathrm{ps}}, \bar{\omega}_{\mathrm{cs}})$, its partial derivatives may be expressed as

$$\frac{\partial X_{ij}^s}{\partial\boldsymbol{Y}} = \frac{\partial X_{ij}^s}{\partial N_{\perp}}\frac{\partial N_{\perp}}{\partial\boldsymbol{Y}} + \frac{\partial X_{ij}^s}{\partial N_{\parallel}}\frac{\partial N_{\parallel}}{\partial\boldsymbol{Y}} + \frac{\partial X_{ij}^s}{\partial\beta_{Ts}}\frac{\partial\beta_{Ts}}{\partial\boldsymbol{Y}} + \frac{\partial X_{ij}^s}{\partial\hat{\omega}_{\mathrm{ps}}}\frac{\partial\hat{\omega}_{\mathrm{ps}}}{\partial\boldsymbol{Y}} + \frac{\partial X_{ij}^s}{\partial\hat{\omega}_{\mathrm{cs}}}\frac{\partial\hat{\omega}_{\mathrm{cs}}}{\partial\boldsymbol{Y}}$$

where the partial derivatives of the susceptibility tensor depend only upon the plasma dispersion model while the partial derivatives with respect to $\boldsymbol{Y}$ are functions of the toroidal MHD equilibrium and the coordinate system only. Taking advantage of chain derivatives, the structure of the ray-tracing formulation becomes highly modular as a consequence of the separation between the physics and the metrics. Such modularity is present in the ray-tracing code $C3PO$ by construction.

3. Curvilinear coordinate representation

The ray-tracing code is primarily designed to calculate rf wave propagation in toroidal magnetized plasmas with nested magnetic flux surfaces. The plasma magnetic field $\boldsymbol{B} = \bar{\boldsymbol{B}} + \tilde{\boldsymbol{B}}$ is decomposed into an axisymmetric equilibrium component $\bar{\boldsymbol{B}}$ and a perturbation $\tilde{\boldsymbol{B}}$. Similarly, the density and temperature for every species s are decomposed as $n_s = \bar{n}_s + \tilde{n}_s$ and $T_s = \bar{T}_s + \tilde{T}_s$, respectively. There is no specific ordering between equilibrium and perturbation quantities, although it is assumed that the magnetic perturbation remains sufficiently small such that the magnetic structure is preserved[6]. This is valid for drift wave type fluctuations and for the toroidal magnetic ripple as far as it remains a small perturbation of the axisymmetric toroidal MHD equilibrium [26].

As shown in figure 1, a flux coordinate system $(\bar{\psi}, \theta, \phi)$ is defined from the axisymmetric toroidal MHD equilibrium, where θ is the poloidal angle measured counter-clockwise from

[6] Ergodization of the magnetic field lines must be negligible.

the horizontal outboard mid-plane, ϕ is the toroidal angle measured clockwise from the top and $2\pi\bar{\psi}$ is the poloidal magnetic flux according to the following expression of the equilibrium magnetic field [10]

$$\bar{\boldsymbol{B}} = I\left(\bar{\psi}\right)\boldsymbol{\nabla}\phi + \boldsymbol{\nabla}\phi \times \boldsymbol{\nabla}\bar{\psi} \tag{4}$$

which can be rewritten as

$$\bar{\boldsymbol{B}} = \bar{B}_s\hat{s} + \bar{B}_\phi\hat{\phi} = \sigma_I\bar{B}_P\hat{s} + \sigma_B\bar{B}_T\hat{\phi} = \left(\bar{\mathcal{P}}\hat{s} + \bar{\mathcal{T}}\hat{\phi}\right)\bar{B} \tag{5}$$

where σ_B is the sign of $\bar{B}_\phi$ and σ_I is the sign of the toroidal plasma current I_ϕ, $\bar{B} = \|\bar{\boldsymbol{B}}\|$, $\bar{\mathcal{P}} = \bar{B}_s/\bar{B}$ and $\bar{\mathcal{T}} = \bar{B}_\phi/\bar{B}$. Here, $\bar{B}_T = |\mathcal{I}(\bar{\psi})|/R$ with $\mathcal{I}(\bar{\psi})$ the current flux function, $\bar{B}_P = \|\boldsymbol{\nabla}\bar{\psi}\|/R$ are the definite positive amplitudes of the toroidal and poloidal magnetic fields, and R is the local major radius. Here, the vectors $(\hat{\psi}, \hat{s}, \hat{\phi})$ form a local orthogonal basis such that $\hat{s} = \hat{\phi} \times \hat{\psi}$ with $\boldsymbol{\nabla}\bar{\psi} = \sigma_I\|\boldsymbol{\nabla}\bar{\psi}\|\hat{\psi}$, $\hat{s}$ being orientated counter-clockwise in the poloidal plane.

It is convenient to introduce a radial coordinate $\bar{\rho}$ that is a monotonic flux function going from 0 at the center to 1 at the edge. It is defined as $\bar{\rho}(\bar{\psi}) = r(\bar{\psi}, 0)/a$ where r is the distance between the magnetic axis and the position $(\bar{\psi}, \theta)$, $a = r(\bar{\psi}_a, 0)$, as shown in figure 1 and $\bar{\psi}_a$ is the poloidal magnetic flux at the separatrix. Even if the coordinate system $(\bar{\rho}, \theta, \phi)$ is naturally designed to study ray dynamics inside the separatrix, it can be extended beyond the limit $\bar{\rho} > 1$. From the relation $\boldsymbol{\nabla}\bar{\rho} = (\mathrm{d}\bar{\rho}/\mathrm{d}\bar{\psi})\|\boldsymbol{\nabla}\bar{\psi}\|\hat{\psi}$, it is deduced that $\|\boldsymbol{\nabla}\bar{\rho}\| = \partial r/\partial\bar{\psi}|_{\theta=0}R\bar{B}_P/a$ where $\partial r/\partial\psi|_{\theta=0}$ is obtained from the magnetic equilibrium. By construction, $\hat{\rho} = \hat{\psi}$, and the angle α defined by $\hat{\rho}\cdot\hat{r} = \cos\alpha$ and $\hat{s}\cdot\hat{r} = \sin\alpha$ is also introduced, where $\hat{r} = \cos\theta\hat{R} + \sin\theta\hat{Z}$, $\hat{R}$ and $\hat{Z}$ being the normalized vectors along the major axis and the vertical direction of the toroidal configuration. Since $(\hat{r}, \hat{\boldsymbol{\theta}}, \hat{\phi})$ form also a local orthogonal basis, the useful relations are obtained, $\hat{s}\cdot\hat{r} = -\hat{\rho}\cdot\hat{\theta} = \sin\alpha$ and $\hat{s}\cdot\hat{\theta} = \cos\alpha$. For circular and concentric magnetic flux surfaces, $\alpha = 0$.

The perturbation of the magnetic field is expressed as

$$\tilde{\boldsymbol{B}} = \tilde{B}_\rho\hat{\rho} + \tilde{B}_s\hat{s} + \tilde{B}_\phi\hat{\phi} = \left(\tilde{\mathcal{R}}\hat{\rho} + \tilde{\mathcal{P}}\hat{s} + \tilde{\mathcal{T}}\hat{\phi}\right)\tilde{B} \tag{6}$$

with $\tilde{B} = \|\tilde{\boldsymbol{B}}\|, \tilde{\mathcal{R}} = \tilde{B}_\rho/\tilde{B}, \tilde{\mathcal{P}} = \tilde{B}_s/\tilde{B}$ and $\tilde{\mathcal{T}} = \tilde{B}_\phi/\tilde{B}$. Since the components of $\tilde{\boldsymbol{B}}$ for the toroidal magnetic field ripple are usually expressed in the local orthogonal basis $(\hat{R}, \hat{Z}, \hat{\phi})$, the relationship between $(\tilde{B}_\rho, \tilde{B}_s)$ and $(\tilde{B}_R, \tilde{B}_Z)$ is

$$\begin{aligned}\tilde{B}_\rho &= \tilde{B}_R\cos(\theta-\alpha) + \tilde{B}_Z\sin(\theta-\alpha)\\ \tilde{B}_s &= -\tilde{B}_R\sin(\theta-\alpha) + \tilde{B}_Z\cos(\theta-\alpha)\end{aligned} \tag{7}$$

where the angle α is itself a function of $\bar{\rho}$ and θ.

The direction of the magnetic field $\hat{\boldsymbol{b}} = \boldsymbol{B}/\|\boldsymbol{B}\|$ is expressed as $\hat{\boldsymbol{b}} = \mathcal{R}\hat{\rho} + \mathcal{P}\hat{s} + \mathcal{T}\hat{\phi}$ where $\mathcal{R}[\mathcal{P}, \mathcal{T}] = \bar{\mathcal{R}}[\bar{\mathcal{P}}, \bar{\mathcal{T}}]\tilde{B}_n + \tilde{\mathcal{R}}[\bar{\mathcal{P}}, \bar{\mathcal{T}}]\tilde{B}_n$, $\bar{B}_n = \bar{B}/B$, $\tilde{B}_n = \tilde{B}/B$ and $B = \|\boldsymbol{B}\|$, and by definition, $\mathcal{R}^2 + \mathcal{P}^2 + \mathcal{T}^2 = 1$. For magnetic perturbations such that $\tilde{\mathcal{R}} = 0$, $\mathcal{R} = 0$ since $\bar{\mathcal{R}} = 0$.

For the coordinate system $(\bar{\rho}, \theta, \phi)$, the covariant basis is

$$\boldsymbol{e}_i = \left(\frac{\hat{r}}{\|\boldsymbol{\nabla}\rho\|\cos\alpha}, \frac{r\hat{s}}{\cos\alpha}, R\hat{\phi}\right) \tag{8}$$

and the contravariant one is

$$\boldsymbol{e}^j = \left(\|\boldsymbol{\nabla}\rho\|\,\hat{\boldsymbol{\rho}}, \frac{\hat{\theta}}{r}, \frac{\hat{\phi}}{R}\right) \tag{9}$$

such that covariant components $k_i = (k_\rho, m, n)$ of the wave vector $\boldsymbol{k}$, which are canonically conjugate to $(\bar{\rho}, \theta, \phi)$, are

$$k_\rho = \frac{\hat{r}\cdot\boldsymbol{k}}{\|\boldsymbol{\nabla}\bar{\rho}\|\cos\alpha} \tag{10}$$

$$m = \frac{r\hat{s}\cdot\boldsymbol{k}}{\cos\alpha} \tag{11}$$

$$n = R\hat{\phi}\cdot\boldsymbol{k} \tag{12}$$

where (m, n) are the usual poloidal and toroidal mode numbers, respectively.

From (1) and (2), the explicit ray equations in canonical coordinates become

$$\frac{\mathrm{d}\bar{\rho}}{\mathrm{d}t} = -\frac{\partial\mathcal{D}}{\partial k_\rho}\Big/\frac{\partial\mathcal{D}}{\partial\omega}, \qquad \frac{\mathrm{d}\theta}{\mathrm{d}t} = -\frac{\partial\mathcal{D}}{\partial m}\Big/\frac{\partial\mathcal{D}}{\partial\omega},$$
$$\frac{\mathrm{d}\phi}{\mathrm{d}t} = -\frac{\partial\mathcal{D}}{\partial n}\Big/\frac{\partial\mathcal{D}}{\partial\omega} \tag{13}$$

and

$$\frac{\mathrm{d}k_\rho}{\mathrm{d}t} = \frac{\partial\mathcal{D}}{\partial\bar{\rho}}\Big/\frac{\partial\mathcal{D}}{\partial\omega}, \qquad \frac{\mathrm{d}m}{\mathrm{d}t} = \frac{\partial\mathcal{D}}{\partial\theta}\Big/\frac{\partial\mathcal{D}}{\partial\omega},$$
$$\frac{\mathrm{d}n}{\mathrm{d}t} = \frac{\partial\mathcal{D}}{\partial\phi}\Big/\frac{\partial\mathcal{D}}{\partial\omega}. \tag{14}$$

For an axisymmetric magnetic configuration, $\partial\mathcal{D}/\partial\phi = 0$, and n is invariant along the ray trajectory. In this case, only the first five equations need to be solved. However, in the presence of plasma fluctuations or when the toroidal magnetic ripple is considered, the toroidal mode number evolves along the ray trajectory and must be calculated like any other coordinate.

4. Initial and boundary conditions

4.1. Wave parameters at launch

Generally, the ray initial conditions are determined at the starting location $\boldsymbol{X}_0 = \{\bar{\rho}_0, \theta_0, \phi_0\}$ with spectral parameters $m_0\,[n_0]$ and $k_{\|0} = \boldsymbol{k}_0\cdot\boldsymbol{B}$ for a tokamak [reverse field pinch] magnetic configuration obtained from the launching structure properties. With $k_{\perp 0}$ calculated from the dispersion relation, $n_0\,[m_0]$ and $k_{\rho 0}$ must be expressed as a function of $m_0\,[n_0]$, $k_{\|0}$ and $k_{\perp 0}$.

Knowing that

$$k_\| = \boldsymbol{k}\cdot\hat{\boldsymbol{b}} = k_\rho\,\|\boldsymbol{\nabla}\bar{\rho}\|\,\mathcal{R} + \lambda\frac{m}{r} + \frac{n}{R}\mathcal{T} \tag{15}$$

where $\lambda = (\mathcal{P}\cos\alpha - \mathcal{R}\sin\alpha)$ and $k^2 = \|\,\boldsymbol{k}\,\|^2 = k_\rho^2\|\boldsymbol{\nabla}\bar{\rho}\|^2 - 2k_\rho\|\boldsymbol{\nabla}\bar{\rho}\|m\sin\alpha/r + m^2/r^2 + n^2/R^2$, the value of $k_\perp^2 = k^2 - k_\|^2$ is

$$\begin{aligned}k_\perp^2 = {}& k_\rho^2\,\|\boldsymbol{\nabla}\bar{\rho}\|^2\left(1-\mathcal{R}^2\right) + \frac{m^2}{r^2}\left(1-\lambda^2\right)\\ & - 2\frac{m}{r}\left[k_\rho\,\|\boldsymbol{\nabla}\bar{\rho}\|\sin\alpha + \lambda\left(k_\rho\,\|\boldsymbol{\nabla}\bar{\rho}\|\,\mathcal{R} + \frac{n}{R}\mathcal{T}\right)\right]\\ & + \frac{n^2}{R^2}\left(1-\mathcal{T}^2\right) - 2k_\rho\,\|\boldsymbol{\nabla}\bar{\rho}\|\,\frac{n}{R}\mathcal{R}\mathcal{T}.\end{aligned} \tag{16}$$

From the initial values of m_0, $k_{\|0}$ and $k_{\perp 0}^2$, the toroidal number at launch is thus

$$n_{0\mp}^{\rm tok} = R\left[\mathcal{T}\left(k_\| - \frac{m_0}{r}\mathcal{P}\cos\alpha\right) \mp \sqrt{\Delta^{\rm tok}}\right] / \left(1-\mathcal{P}^2\right) \quad (17)$$

for a tokamak configuration, with

$$\begin{aligned}\Delta^{\rm tok} = {} & \mathcal{T}^2\left(k_{\|0} - \frac{m_0}{r}\mathcal{P}\cos\alpha\right)^2 \\ & - (1-\mathcal{P}^2)\left[\left(k_{\|0} - \frac{m_0}{r}\lambda\right)^2(1-\mathcal{R}^2)\right. \\ & + \frac{m_0^2}{r^2}(1-\lambda^2)\mathcal{R}^2 - 2\frac{m_0}{r}\left(k_{\|0} - \frac{m_0}{r}\lambda\right) \\ & \left.\times(\sin\alpha + \lambda\mathcal{R})\,\mathcal{R} - k_{\perp 0}^2\mathcal{R}^2\right] \end{aligned} \quad (18)$$

and the radial component of the wave vector $k_{\rho 0,\pm}^{\rm tok}$ is directly derived from (15)[7].

Assuming $\mathcal{R} = 0$ for the reverse field pinch magnetic configuration, the initial poloidal mode number is $m_0^{\rm rfp} = r(k_{\|0} - n_0\mathcal{T}/R)/(\mathcal{P}\cos\alpha)$ and the radial wave vector component is $k_{\rho 0,\pm}^{\rm rfp} = (m_0\sin\alpha/r \pm \sqrt{\Delta^{\rm rfp}})/\|\boldsymbol{\nabla}\bar\rho\|$ with $\Delta^{\rm rfp} = k_{\perp 0}^2 + k_{\|0}^2 - m_0^2\cos^2\alpha r^2 - n_0^2/R^2$.

Both $k_{\rho 0,\pm}^{\rm tok}$ and $k_{\rho 0,\pm}^{\rm rfp}$ have two solutions corresponding to inward and outward propagating waves. The solution is chosen such that the ray initially propagates inward, meaning $\mathrm{d}\bar\rho/dt < 0$. For positive energy waves with $\partial\mathcal{D}/\partial\omega > 0$ and using $d\bar\rho/dt < 0$, the condition of inward propagation is

$$\begin{aligned}\frac{\|\boldsymbol{\nabla}\bar\rho\|}{k_\perp}\left(k_\rho\,\|\boldsymbol{\nabla}\bar\rho\|\left(1-\mathcal{R}^2\right) - \frac{m}{r}\left(\sin\alpha + \mathcal{R}\lambda\right) - \frac{n}{R}\mathcal{R}\mathcal{T}\right) \\ \times\frac{\partial\mathcal{D}}{\partial k_\perp} + \|\boldsymbol{\nabla}\bar\rho\|\,\mathcal{R}\frac{\partial\mathcal{D}}{\partial k_\|} < 0\end{aligned} \quad (19)$$

and if $\mathcal{R}$ is a small perturbation of the axisymmetric toroidal MHD equilibrium, the condition (19) reduces to

$$\frac{\|\boldsymbol{\nabla}\bar\rho\|}{k_\perp}\left(k_\rho\,\|\boldsymbol{\nabla}\bar\rho\| - \frac{m}{r}\sin\alpha\right)\frac{\partial\mathcal{D}}{\partial k_\perp} < 0. \quad (20)$$

In general, this condition is satisfied, in which case (20) takes the simple form $k_\rho\partial\mathcal{D}/\partial N_\perp < 0$ or equivalently $k_\rho S_\perp < 0$ where $S_\perp$ is the energy flow density in the perpendicular direction.

4.2. Ray trajectories at the plasma edge

Depending upon the type of rf wave, rays may leave the plasma or encounter a cut-off layer. Hence, since the EC wave can propagate continuously from the magnetized plasma to the vacuum, calculations are simply stopped when $\bar\rho > 1$. From that point, it is possible to calculate analytically the straight line trajectory of the wave until it hits the first wall of the vacuum chamber. A model of interaction between the wall and the EC wave can be used to evaluate the fraction of power that is reflected or back-scattered. In this case, a single ray leaving the plasma will split into multiple rays that re-enter the plasma with different polarizations, each of them carrying a fraction of the incident power. Such a method has been recently implemented successfully in *C3PO*, for describing EC wave propagation when the single pass power absorption is very weak [27].

The situation is different for the LH wave since it does not propagate in vacuum. Indeed, a cut-off near the plasma edge prevents the wave from leaving the plasma [12]. When the cut-off layer lies inside the separatrix, the reflection takes place naturally during the propagation and the ray turns back toward the core of the plasma. Even if $k_\rho \to 0$ at the cut-off, the WKB approximation does not fail since $k_\| \neq 0$[8]. Consequently, the direction of the energy flow $\boldsymbol{S}$ calculated by ray tracing remains valid after the cut-off. This layer acts as a perfect plane mirror for the LH wave and the single assumption done in the calculations is that no power is transmitted in the evanescent region by mode conversion. However, it is possible to calculate the fraction of power that is lost by this process by solving the local full-wave equation [20, 28]. Such an effect could contribute to lower the amount of LH power available for current drive when plasma absorption is very weak, such that multiple passes are necessary for the wave to be damped.

If the cut-off layer is located outside the separatrix, i.e. $\bar\rho > 1$, a specular reflection

$$\check{\boldsymbol{k}}\cdot\nabla\rho = -\boldsymbol{k}\cdot\nabla\rho \quad (21)$$

$$\check{\boldsymbol{k}}\cdot\hat{\boldsymbol{b}} = \boldsymbol{k}\cdot\hat{\boldsymbol{b}} \quad (22)$$

$$\check{\boldsymbol{k}}\cdot\left(\nabla\rho\times\hat{\boldsymbol{b}}\right) = \boldsymbol{k}\cdot\left(\nabla\rho\times\hat{\boldsymbol{b}}\right) \quad (23)$$

is enforced at $\bar\rho = 1$ [29]. When $\nabla\rho\cdot\boldsymbol{B} = 0$, the conservation of $k_\|$ and $k_\perp$ from (21)–(23) guarantees that $\mathcal{D}(\check{\mathcal{N}}, \beta_{Ts}, \bar\omega_{\rm ps}, \bar\omega_{\rm cs}) = 0$, where $\check{\mathcal{N}} = c\check{\boldsymbol{k}}/\omega$. This is the case for axisymmetric MHD equilibrium. When the toroidal magnetic ripple is considered, $\check{k}_\perp \simeq k_\perp$ at the reflection, since $\nabla\rho\cdot\boldsymbol{B} = \mathcal{O}(\nabla\tilde\rho\cdot\tilde{\boldsymbol{B}})$ as ρ is the solution of an approximate equation (see section 7.1, equation (40)). As a consequence, at each reflection, $\mathcal{D}(\check{\mathcal{N}}, \beta_{Ts}, \bar\omega_{\rm ps}, \bar\omega_{\rm cs})$ may exhibit a jump, since $\mathcal{D}$ is a very sensitive function. In order to reduce cumulative errors at these positions of the ray, the dispersion relation must be therefore locally solved for $\check{k}_\perp$ before continuing the calculation of the ray trajectory.

5. Numerical algorithm

5.1. Toroidal MHD equilibrium interpolation

Since toroidal MHD equilibrium solvers usually give field components on a discrete two-dimensional grid mesh $(\bar\rho, \theta)$, it is necessary to perform a numerical interpolation at the ray location[9] [9]. Among several methods[10], the combination

[7] For an axisymmetric magnetic configuration, $\mathcal{R} = 0$ and $\Delta^{\rm tok} = 0$.

[8] The curvature radius of the cut-off layer must nevertheless be much larger than the wavelength in the parallel direction $\lambda_\|$, a condition that is generally satisfied.

[9] When field components are given on a discrete (R, Z) two-dimensional grid mesh, data resampling must be first performed to express them on the discrete two-dimensional grid mesh $(\bar\rho, \theta)$. It is the case for EFIT code [30].

[10] Note that this method is universal, and may be used for any numerical data provided from a Grad–Shafranov equation solver. However, for some codes that use finite elements techniques, it may be useful to use those directly instead of performing an additional interpolation procedure based on a spline-Fourier expansion. Indeed, finite elements contain all data needed for calculating

of Fourier expansion and spline interpolation is appropriate when $(\bar{\rho}, \theta, \phi)$ coordinates are used, taking into account the periodicity of the poloidal angle θ. Any function of the form $g(\bar{\rho}, \theta)$ may be therefore expressed as a Fourier sum

$$g(\bar{\rho}, \theta) = a_0(\bar{\rho}) + \sum_{p=1}^{p=N_p} a_p^c(\bar{\rho}) \cos(p\theta) + a_p^s(\bar{\rho}) \sin(p\theta) \tag{24}$$

where $N_p \leqslant N_\theta/2$ according to Shannon's theorem. Here, N_θ is the number of discrete poloidal values [14]. The Fourier coefficients $a_p^{(c,s)}(\bar{\rho}_i)$ are determined on each discrete magnetic flux surface ρ_i in order to satisfy the condition

$$g(\bar{\rho}_i, \theta_j) = a_{0,i} + \sum_{p=1}^{p=N_p} a_{p,i}^c \cos(p\theta_j) + a_{p,i}^s \sin(p\theta_j) \tag{25}$$

and the coefficients $a_p^{(c,s)}(\bar{\rho})$ at any radial position $\bar{\rho}_i < \bar{\rho} < \bar{\rho}_{i+1}$ are obtained using a usual cubic spline interpolation $a_p^{(c,s)}(\bar{\rho}) = \alpha_{p,i}^{(c,s)} \bar{\rho}^3 + \beta_{p,i}^{(c,s)} \bar{\rho}^2 + \gamma_{p,i}^{(c,s)} \bar{\rho} + \delta_{p,i}^{(c,s)}$, where the third order polynomial coefficients are calculated such that at all nodes $\bar{\rho}_i$, $a_p^{(c,s)}(\bar{\rho})$ is continuous, as well as its first and second derivatives. It is then straightforward to calculate at any $(\bar{\rho}, \theta)$ position all the partial derivatives used by the ray tracing.

For density and temperature profiles, only a cubic spline interpolation is performed, since these parameters are functions of $\bar{\rho}$ only.

The numbers of Fourier modes are chosen such that the relative accuracy at all grid points of the discrete numerical equilibrium never exceeds 10^{-4}, even near a X-point, if it exists, where the local curvature radius of magnetic flux surfaces becomes very small. If smoothing is needed because the original equilibrium data are ragged, it is possible to use fewer coefficients in the Fourier expansion series.

5.2. Ordinary differential equation solver

Relations (13) form a system of first order coupled differential equations. Knowing the initial wave characteristics $(k_{\rho 0}, m_0, n_0)$ at the position $(\bar{\rho}_0, \theta_0, \phi_0)$, the objective is to evaluate (k_ρ, m, n) and $(\bar{\rho}, \theta, \phi)$ at all values along the ray parametrized by a time-like parameter t.

There are several algorithms for solving this initial value problem, minimizing propagation of the numerical error. One of the most popular is the adaptive stepsize Runge–Kutta algorithm which belongs to the Euler–Cauchy algorithm family [14]. Among other methods, the 'predictor–corrector' which is a robust technique belonging to the general Adams algorithm family has been used also extensively for solving ray-tracing equations [25]. As to solve the equations of rays consists in determining the trajectory within the phase space subject to the condition $\mathcal{D} = 0$, where $\mathcal{D}$ acts as a Hamiltonian operator, symplectic integration may be employed for this purpose [31]. With this numerical method, the constraint $\mathcal{D} = 0$ is exactly satisfied, while it is only fulfilled approximately with the Runge–Kutta or 'predictor–corrector' algorithms.

Nevertheless, taking into consideration its facility to be implemented and its numerical efficiency, the adaptive stepsize Runge–Kutta–Fehlberg algorithm also known as Runge–Kutta (4, 5) pair has been implemented in *C3PO* for integrating ray equations [15, 32]. It uses a Runge–Kutta method of fifth order to estimate the local error in a Runge–Kutta method of fourth order. Working with a predetermined global error, the initial step size is modified along the trajectory if the error at a given step is not within the required bounds. A nice feature of this technique is that it reduces the global number of evaluations with an integrator that can reuse some of the same values in both the fourth order and fifth order methods. Therefore, only six evaluations are required at each step. Provided that the stability of the fifth order formula is acceptable, advancing the integration with the higher-order result leads to a more accurate integration at no additional cost.

6. Code validation

6.1. Wave propagation in vacuum

In the development of a code, the verification and benchmarking process is a critical step to demonstrate that the numerical tool meets its specified functionality. In order to validate the implementation of the curvilinear coordinate system, wave propagation in vacuum is first considered. In this case, electromagnetic wave trajectories are straight lines. At $\bar{\rho} = 1$ specular reflection is enforced, such that the ray remains confined in the machine. Circular concentric poloidal magnetic flux surfaces are considered with $R_0 = 3.0\,\mathrm{m}$ and $a = 0.9\,\mathrm{m}$. The toroidal magnetic field on axis is $B_{T0} = 3\,\mathrm{T}$ and the poloidal magnetic field is considered to be created by a virtual toroidal current I_p of 1.0 MA, its radial profile being defined by a safety factor of the form $q(\bar{\rho}) = q_{\min} + (q_{\max} - q_{\min})\bar{\rho}^2$ with $q_{\min} = 1$ and $q_{\max}$ obtained from Ampère's law.

Since rays are straight lines, it is possible to analytically calculate the exact ray trajectories. As $\|\nabla\bar{\rho}\| = 1$, the initial value of the wave vector $\boldsymbol{k}_0 = k_{\rho 0}\hat{\rho} + k_{\theta 0}\hat{\theta} + k_{\phi 0}\hat{\phi}$ with $k_{\theta 0} = m_0/(a\bar{\rho}_0)$ and $k_\phi = n_0/R_0$ is expressed in Cartesian coordinates using the relation $\boldsymbol{k}_0^{\mathrm{cart.}} = \mathbb{M}_{cc}(\theta_0, \phi_0) \cdot \boldsymbol{k}_0^{\mathrm{curv.}}$ where

$$\mathbb{M}_{cc}(\theta, \phi) = \begin{pmatrix} \cos\theta\cos\phi & -\sin\theta\cos\phi & -\sin\phi \\ \cos\theta\sin\phi & -\sin\theta\sin\phi & \cos\phi \\ \sin\theta & \cos\theta & 0 \end{pmatrix}. \tag{26}$$

The initial position at launch in the Cartesian coordinates $(\bar{x}_0, \bar{y}_0, \bar{z}_0)$ normalized to a_p is given by the relations

$$\begin{aligned} \bar{x}_0 &= \left(\frac{R_p}{a_p} + \bar{\rho}_0 \cos\theta_0\right) \cos\phi_0,\ \bar{y}_0 \\ &= \left(\frac{R_p}{a_p} + \bar{\rho}_0 \cos\theta_0\right) \sin\phi_0,\ \bar{z}_0 = \bar{\rho}_0 \sin\theta_0 \end{aligned} \tag{27}$$

where $(\bar{\rho}_0, \theta_0, \phi_0)$ are initial curvilinear coordinates. Since the ray is a straight line, its position is given by the parametric equation

$$\begin{aligned} \bar{x}^{(l)} &= \bar{x}_0 + k_{x0} s(l)/k_0,\ \bar{y}^{(l)} = \bar{y}_0 + k_{y0} s(l)/k_0,\ \bar{z}^{(l)} \\ &= \bar{z}_0 + k_{z0} s(l)/k_0 \end{aligned} \tag{28}$$

magnetic field values at any radial and poloidal positions including first and second derivatives with an excellent accuracy. The choice to use the spline-Fourier expansion is here justified by the need to have a ray-tracing solver as general as possible.

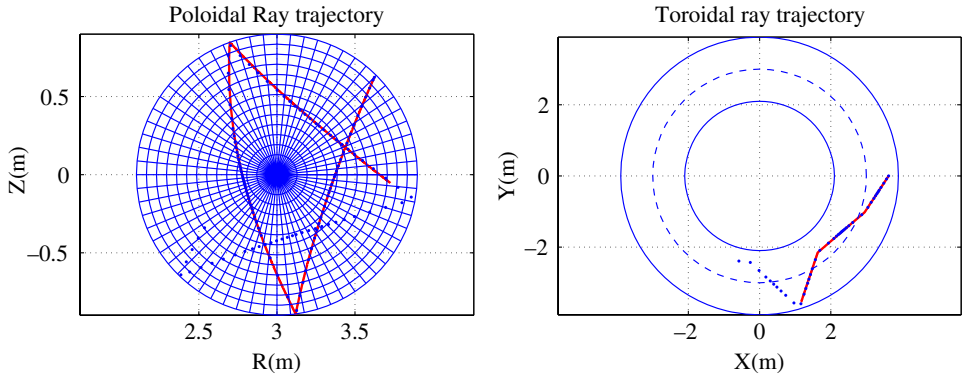


Figure 2. Poloidal (left) and toroidal (right) projections of the ray trajectory for an electromagnetic wave in vacuum. Full red lines correspond to the ray tracing and dashed blue ones to the straight line model.

where $s(l)$ is the ray length starting from $s(l=0)=0$ at $(\bar{x}_0, \bar{y}_0, \bar{z}_0)$. Once $(\bar{x}^{(l)}, \bar{y}^{(l)}, \bar{z}^{(l)})$ is calculated, $(\bar{\rho}, \theta, \phi)$ is obtained by inverting (27). From the determination of $(\bar{\rho}, \theta, \phi)$, it is possible to evaluate $\boldsymbol{k}_l^{\rm curv}$ at all positions, using $\boldsymbol{k}_l^{\rm cart} = \boldsymbol{k}_0^{\rm cart}$ and $\boldsymbol{k}_l^{\rm curv} = \mathbb{M}_{cc}^{-1}(\theta,\phi)\cdot\boldsymbol{k}_0^{\rm cart}$. Then $m^{(l)} = k_\theta^{(l)}\bar{\rho}^{(l)}a_p$ and $n^{(l)} = k_\phi^{(l)}R^{(l)}$.

In order to keep the ray inside the toroidal chamber, a specular reflection is enforced at $\bar{\rho}=1$. If the condition $\bar{\rho}^{(l)}>1$ is encountered, then the new direction of the $i+1$ ray segment $\boldsymbol{k}_{0(i+1)}^{\rm cart}$ is derived from $\boldsymbol{k}_{0(i)}^{\rm cart}$ according to the conditions $\boldsymbol{k}_{0(i+1)}^{\rm cart}\cdot\hat{\rho}^{(l-1)} = -\boldsymbol{k}_{0(i)}^{\rm cart}\cdot\hat{\rho}^{(l-1)}$, $\boldsymbol{k}_{0(i+1)}^{\rm cart}\cdot\hat{\theta}^{(l-1)} = \boldsymbol{k}_{0(i)}^{\rm cart}\cdot\hat{\theta}^{(l-1)}$ and $\boldsymbol{k}_{0(i+1)}^{\rm cart}\cdot\hat{\phi}^{(l-1)} = \boldsymbol{k}_{0(i)}^{\rm cart}\cdot\hat{\phi}^{(l-1)}$ where $(\hat{\rho},\hat{\theta},\hat{\phi})$ are evaluated at the increment $(l-1)$. It corresponds to the relation

$$\boldsymbol{k}_{0(i+1)}^{\rm cart} = [\mathbb{M}_{cc1}^{-1}(\theta^{(l-1)},\phi^{(l-1)})\cdot\mathbb{M}_{cc2}(\theta^{(l-1)},\phi^{(l-1)})]\cdot\boldsymbol{k}_{0(i)}^{\rm cart} \tag{29}$$

with

$$\mathbb{M}_{cc1}(\theta,\phi) = \begin{pmatrix} \cos\theta\cos\phi & \cos\theta\sin\phi & \sin\theta \\ -\sin\theta\cos\phi & -\sin\theta\sin\phi & \cos\theta \\ \sin\phi & -\cos\phi & 0 \end{pmatrix} \tag{30}$$

and

$$\mathbb{M}_{cc2}(\theta,\phi) = \begin{pmatrix} -\cos\theta\cos\phi & -\cos\theta\sin\phi & -\sin\theta \\ -\sin\theta\cos\phi & -\sin\theta\sin\phi & \cos\theta \\ \sin\phi & -\cos\phi & 0 \end{pmatrix}. \tag{31}$$

In the example given here, the ray is launched from the position $\bar{\rho}_0 = 0.98$, $\theta_0 = \pi/4$ and $\phi_0 = 0$, with spectral properties $\omega/(2\pi) = 530.15\,\text{GHz}$, $N_{\|0} = 0.5$ and $m_0 = -3000$ such that the wave can propagate into the vacuum region. As expected, the ray trajectory is made of pieces of straight lines, which can be easily seen from the toroidal projection in figure 2. Because of the toroidal topology, the poloidal projection in figure 2 does not give straight lines. This result highlights the fact that this type of representation can be misleading. Analytical and numerical results agree perfectly well.

6.2. *Numerical accuracy and interpolation method*

An exact comparison between results obtained with *C3PO* and *FRTC* codes has been carried out for the LH wave in a analytically defined toroidal MHD equilibrium using the cold plasma model [33]. Here again, poloidal magnetic flux surfaces are concentric and circular. MHD equilibrium parameters are $R_0 = 3.05\,\text{m}$, $a = 0.95\,\text{m}$, while the toroidal magnetic field on axis is $\bar{B}_{\rm T0} = 3.2\,\text{T}$ and the toroidal current is $I_p = 3.5\,\text{MA}$. The current density profile is parabolic $j = 2I_P(1-\bar{\rho}^2)/(\pi a^2)$ and the corresponding poloidal magnetic field on axis is $\bar{B}_{P0} = \mu_0 I_P[1-(1-\bar{\rho}^2)^2]/(2\pi a\bar{\rho})$. The toroidal [poloidal] magnetic fields are given by the relations $\bar{B}_{T[P]} = \bar{B}_{T[P]0}R_0/R$, and the the poloidal magnetic flux is $\bar{\psi} = \mu_0 I_P R_0\bar{\rho}^2[1-\bar{\rho}^2/4]/(2\pi)$. The density profile in this hydrogen plasma is $n(\bar{\rho}) = (n_0-n_a)(1-\bar{\rho}^2)+n_a$ with $n_0 = 5.0\times10^{19}\,\text{m}^{-3}$ and $n_a = 1.0\times10^{17}\,\text{m}^{-3}$.

The ray is launched in the slow mode from the position $\bar{\rho}_0 = 0.968$, $\theta_0 = \phi_0 = 0$ with the spectral properties $\omega/(2\pi) = 3.7\text{GHz}$, $N_{\phi 0} = -1.8$ and $m_0 = 0$ as indicated in [33]. If $\boldsymbol{k}$ is expressed in coordinates (r,θ,ϕ) as $\boldsymbol{k} = k_r\hat{r} + k_\theta\hat{\theta} + k_\phi\hat{\phi}$ where

$$k_\rho = \frac{k_r}{\|\boldsymbol{\nabla}\bar{\rho}\|\cos\alpha} \tag{32}$$

$$m = r(k_\theta + k_r\tan\alpha) \tag{33}$$

$$n = Rk_\phi \tag{34}$$

the initial toroidal mode number is $n_0 = \omega R_0 N_{\phi 0}/c$, and the corresponding initial parallel index of refraction is $N_{\|0} = \bar{T}N_{\phi 0}$.

The numerical toroidal MHD equilibrium is defined on a numerical grid of size (1001×65) in the radial and poloidal directions, respectively. The vectorization procedure of the toroidal MHD equilibrium takes about 5.5 s for 32 Fourier modes on a computer equipped with a 64 bits biprocessor INTEL Core i7 clocked at 2.66 GHz, with 8 GBytes of RAM memory, whatever the interpolating method. Ray calculation parameters for the example presented in figure 3 are typical of tokamak current drive simulation. They are stored in the memory each time the radial increment $\Delta\bar{\rho}$ reaches 10^{-4}. $\Delta\bar{\rho}$ may be adapted depending upon the problem. In the Runge–Kutta–Fehlberg method, the generalized incremental step h is adjusted such that the norm of the variation along the six directions of extrapolation never exceeds the tolerance level $h_{\max}$, which is set to 10^{-12}. The norm is defined as $\|\boldsymbol{X}\|_\infty =$

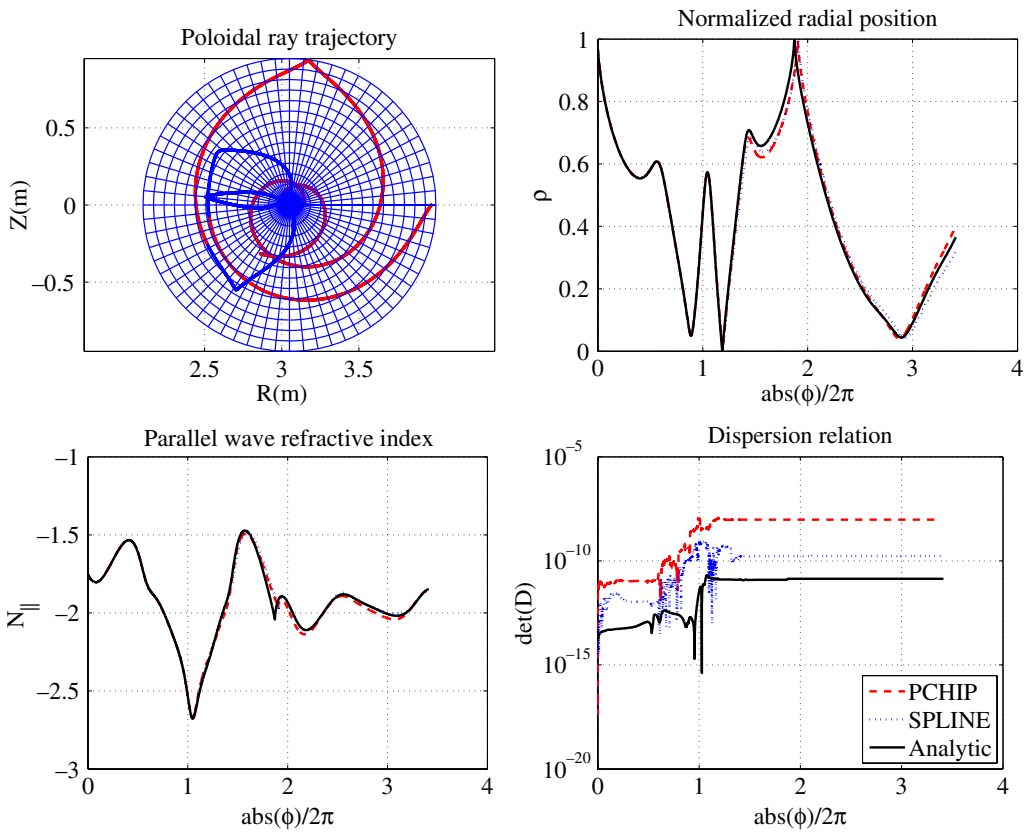


Figure 3. Poloidal projection of the ray trajectory for the LH wave calculated by $C3PO$ with $N_{\phi 0} = -1.8$ (top left). The red curve corresponds to the slow mode and the blue curve corresponds to the fast mode. Calculations are performed with a piecewise cubic Hermite interpolating polynomial for the toroidal MHD equilibrium. Radial positions of the ray trajectory (top right), parallel refractive index (bottom left) and determinant of the dispersion tensor (bottom right) are calculated by three different methods. The full black curve corresponds to an analytical equilibrium, the red dashed and blue dotted ones to numerical equilibrium reconstructed with piecewise cubic Hermite interpolating polynomials and cubic spline, respectively. The progressive divergence between the results appears after the rays passed in the vicinity close to the center of plasma where interpolation error becomes large.

$\max_{i\in[1,N]}(|)$, where N is the dimension of the vector $\boldsymbol{X}$. The level is arbitrarily chosen such that the ray trajectory as well as all wave parameters are independent of its value. Below 10^{-12}, the gain on the ray accuracy is marginal while the computer time increases. The initial time increment is set to $h_0 = 10^{-4}$ in normalized units, and the calculation ends when $t_{\rm end} = 10\,000$, with a maximum of 60 000 values stored in the memory. Since the time step h evolves along the ray trajectory depending upon the ray curvature, the effective number of steps is typically much less than $t_{\rm end}/h_0$. For the current example, $t_{\rm end}$ is reached after 5499 steps. Using the numerical equilibrium, the ray trajectory is calculated in 120 s with piecewise cubic Hermite polynomials, and 19 s with cubic splines interpolation techniques, while it takes only 5 s with the analytic equilibrium. Although using the numerical equilibrium is more time consuming than the analytical equilibrium by at least an order of magnitude, typical calculation times remain at an acceptable level for realistic time-dependent tokamak simulations, especially if simultaneous ray calculations are distributed over several processors. It is worth noting that in the present case, where magnetic flux surfaces are circular and concentric and the inverse aspect ratio is small, only 6 Fourier modes and 21 radial nodes are necessary for describing the toroidal MHD equilibrium, regarding the accuracy of the equilibrium reconstruction. Using such low resolution, the time consumption with the numerical equilibrium drops dramatically and becomes even much lower (<1 s) than for the calculations done with the analytical equilibrium (see section 5.1).

As shown in figure 3, the agreement between results obtained with $C3PO$ and *FRTC* codes is excellent, including the locations where the cold mode conversion occurs between slow and fast modes [33]. Small discrepancies are observed between trajectories calculated with an analytical or a numerical equilibrium after the ray passed in the vicinity close to the plasma center. Indeed, in this small region, the accuracy of the calculations becomes rather poor despite the adaptive stepsize in the Runge–Kutta–Fehlberg method. As a consequence, the solution depends upon the interpolation method. Piecewise cubic Hermite interpolating polynomials are usually preferred since they produce no overshoots and fewer oscillations when the equilibrium data are not smooth.

An important numerical feature of $C3PO$ is the fact that $\mathcal{D}$ remains at a low level along the ray trajectory even if rays are reflected at the LFS edge cut-off. This is an unambiguous indication of the high quality of the numerical implementation of the ray equations, the dispersion relation being extremely sensitive to any small cumulative error. By lowering $h_{\rm max}$ down

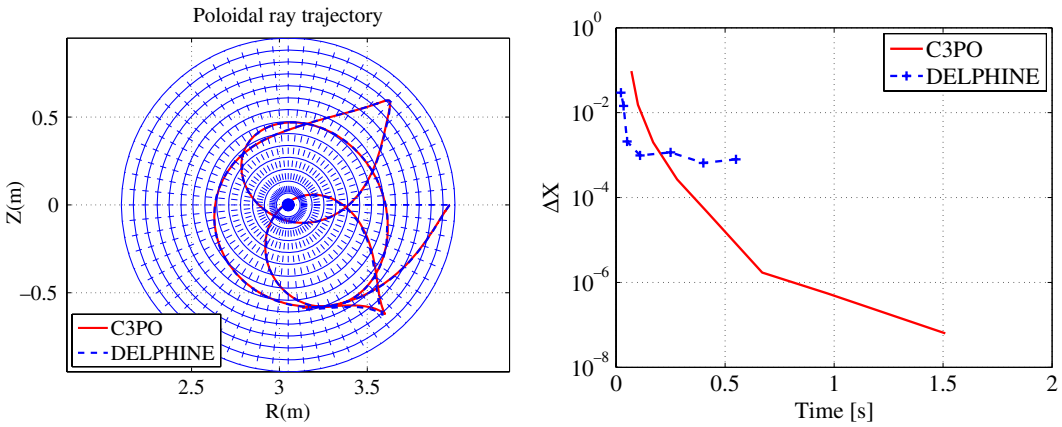


Figure 4. Poloidal projection of the ray trajectory for the LH wave calculated by $C3PO$ (full red curve) and *DELPHINE* ray tracings (dashed blue curve) [5], with $N_{\phi 0} = -2.0$ (left). Distance ΔX normalized to a, between the position at the end of the ray and the corresponding value for the reference case, as a function of the computer computation time, as discussed in section 6.3 (right).

to 10^{-18}, $\mathcal{D}$ never exceeds 10^{-15}, which approaches double-precision accuracy limitation. However, such level of precision requires a longer computing time.

6.3. Comparative computational performances

The advantages of using analytical derivatives over numerical ones in ray-tracing calculations are investigated by comparing performances between $C3PO$ and *DELPHINE* ray-tracing codes, which are based on the same Runge–Kutta procedure for integrating the set of ordinary differential equations [5]. However, in the latter, (R, Z, ϕ) coordinates are used along with two-dimensional local numeric (R, Z) interpolations for calculating ray trajectories. For the benchmarking procedure, the analytically defined toroidal MHD equilibrium introduced in section 6.2 is considered. Wave parameters at launch are also identical, but with $N_{\phi 0} = -2.0$. The interpolated toroidal MHD equilibrium is characterized in $C3PO$ by 21 radial nodes and 6 Fourier modes. Ray trajectories obtained with $C3PO$ and DELPHINE are shown in figure 4, which highlights the good global agreement between the codes.

Since each ray-tracing code has its specific set of internal parameters for optimizing calculations, a universal comparison procedure is performed, which is independent of the numerical method. Code performance is measured at a ray length corresponding to $\Delta\phi \approx 10$ rad approximately, both codes being run on the same computer. By increasing the tolerance level $h_{\max}$ in the Runge–Kutta routine, the calculation time is reduced, but the ray trajectory progressively diverts from a reference path calculated for $C3PO$ with $h^{C3PO}_{\max(\mathrm{ref})} = 10^{-12}$, and for *DELPHINE* with $h^{DELPHINE}_{\max(\mathrm{ref})} = 10^{-12}$. The cumulative errors are characterized by the distance ΔX normalized to a, between the position at the end of the ray and the corresponding value for the reference case. The evolution ΔX with the calculation time is presented in figure 4. By decreasing progressively $h^{C3PO}_{\max}$ from 10^{-4} to 10^{-11}, ΔX improves from $\sim 10^{-1}$ to $\sim 10^{-7}$ with $C3PO$ while the computer time consumption increases from 0.05 to 1.5 s. For the same ray length, such regular behavior is not observed with the *DELPHINE* ray tracing, with $h^{DELPHINE}_{\max}$ ranging from 10^{-4} to 10^{-11}, while the calculation time increases slightly from 0.02 to 0.7 s. The lack of improvement of ΔX with *DELPHINE* over a certain range of tolerances seems to indicate some limits on the accuracy that can be obtained with two-dimensional numerical derivatives. Conversely, significative improvements are still possible with $C3PO$ using analytical derivatives associated with the curvilinear coordinates $(\bar{\rho}, \theta, \phi)$. For the same ΔX value, $C3PO$ and *DELPHINE* calculation times have a similar order of magnitude, as shown in figure 4.

The code comparison between $C3PO$ and *DELPHINE* highlights the advantages of analytical derivatives derived from an interpolated toroidal MHD equilibrium using a spline-Fourier technique over a local numerical (R, Z) interpolation when calculations are performed inside the separatrix, where the magnetic topology corresponds to nested magnetic flux surfaces. However, the approach considered in *DELPHINE* remains fully pertinent outside the separatrix, where the description of the plasma properties is more appropriate with the (R, Z, ϕ) coordinates.

It is worth noting that even if $\Delta\mathbb{X} \leqslant 1$ for $\Delta\phi \approx 10$ rad, it may rapidly approach unity for a longer ray length, a case which is frequently encountered for the LH wave in existing tokamaks when the accessibility is limited or the damping rate is low. Therefore, combining high numerical accuracy with low computing time in ray-tracing codes is essential for heating and current drive calculations in tokamak plasmas.

7. Examples

Four examples are chosen for illustrating the various capabilities of the $C3PO$ ray-tracing code. Firstly, LH wave propagation is calculated in the presence of toroidal magnetic ripple due to the discrete number of coils. It illustrates a case where $\tilde{\mathcal{R}} \neq 0$. Secondly, the case of LH wave propagation in a plasma with fluctuating density near the edge is presented. Next, the multi-wave capacity of the code is illustrated by a standard case of EC wave propagation in ITER. Finally, a case of LH wave propagation calculated in a reverse field pinch (RFP) shows that $C3PO$ is not restricted to tokamaks.

7.1. Effect of toroidal magnetic field ripple on the LH wave

When the discreteness of the magnetic system is accounted for, the toroidal MHD equilibrium is no longer axisymmetric. Fully consistent equilibrium calculations require the use of a 3D equilibrium equation beyond the Grad–Shafranov model [34]. However, since magnetic ripple is typically a very small perturbation of the axisymmetric toroidal MHD equilibrium, the response of the plasma can be generally neglected such that ripple effects can be determined from magnetostatic equations in vacuum [26]. In general, the ripple correction to the magnetic field is expressed in the form of a Fourier series

$$\tilde{B}_{[R,Z,\phi]} = \sum_{j\geqslant 1} \delta^{sj}_{[R,Z]} \sin(jN\phi) + \delta^{cj}_{[R,Z]} \cos(jN\phi) \quad (35)$$

where N is the number of toroidal magnetic field coils and j the harmonic number. Contributions beyond the second harmonic are usually negligible. From (35), it is always possible to calculate $\tilde{B}_{[\rho,s,\phi]}$ using (7).

A simplified model is used for illustrating the effect of magnetic ripple on ray trajectories, where the magnetic flux surfaces are circular and concentric. In that case, the contours of iso-ripple are approximately circles whose center is shifted toward the principal axis of the torus relative to the center of the coils. Knowing the shape of the contours and keeping only the first harmonic contribution, $\tilde{B}_\rho = \delta_\rho(\bar{\rho},\theta)\bar{B}\sin(N\phi)$, $\tilde{B}_\theta = \delta_\theta(\bar{\rho},\theta)\bar{B}\sin(N\phi)$ $\tilde{B}_\varrho = \delta_\rho(\bar{\rho},\theta)\bar{B}\sin(N\phi)$, $\tilde{B}_\theta = \delta_\theta(\bar{\rho},\theta)\bar{B}\sin(N\phi)$ where $R\bar{B} = R_0\bar{B}_0$ and

$$\delta_\rho(\bar{\rho},\theta) = \delta_n\sqrt{\frac{R}{R_0-L_0}}I_1\left(\frac{a_r N}{R_0-L_0}\right) \times \frac{r+L\cos\theta}{\sqrt{r^2+L^2+2rL\cos\theta}} \quad (36)$$

$$\delta_\theta(\bar{\rho},\theta) = \delta_n\sqrt{\frac{R}{R_0-L_0}}I_1\left(\frac{a_r N}{R_0-L_0}\right) \times \frac{L\sin\theta}{\sqrt{r^2+L^2+2rL\cos\theta}} \quad (37)$$

$$\delta_\phi(\bar{\rho},\theta) = \delta_n I_0\left(\frac{a_r N}{R_0-L_0}\right) \quad (38)$$

with $r = \bar{\rho}a$, $a_r \simeq \sqrt{(r^2+L_0^2+2rL_0\cos\theta)(R_0-L_0)/R}$, $a_r \simeq \sqrt{(r^2+L_0^2+2rL_0\cos\theta)(R_0-L_0)/R}$ and $\delta_n = \tilde{F}_0 N/(R_0\bar{B}_0)$. $\tilde{F}_0$ is the perturbation of the magnetic field potential at the center of the coils [26]. In (36)–(38), $I_0(x)$ and $I_1(x)$ are modified Bessel functions of order 0 and 1, respectively. Approximate values of δ_n and L_0 can be obtained from $\delta_\phi \approx 1$ and $L = 0$ at the location of the mean radius of the toroidal windings

Since perturbed magnetic flux surfaces are assumed to be static and nested, fast parallel transport yields $n_s[T_s](\rho(\bar{\rho},\theta,\phi)) = \bar{n}_s[\bar{T}_s](\bar{\rho})$. $n_s[T_s](\rho(\bar{\rho},\theta,\phi)) = \bar{n}_s[\bar{T}_s](\bar{\rho})$ incorporates the effect of the magnetic field perturbation. From the contravariant representation of $\nabla\rho$

$$\nabla\rho = \left(1+\frac{\partial\tilde{\rho}}{\partial\bar{\rho}}\right)\|\boldsymbol{\nabla}\bar{\rho}\|\,\hat{\boldsymbol{\rho}} + \frac{\partial\tilde{\rho}}{\partial\theta}\frac{\hat{\theta}}{r} + \frac{\partial\tilde{\rho}}{\partial\phi}\frac{\hat{\phi}}{R} \quad (39)$$

it is then possible to calculate the explicit form of $\tilde{\rho}(\bar{\rho},\theta,\phi)$ by solving the flux surface equation $\nabla\rho\cdot\boldsymbol{B} = 0$ [4]. Neglecting second order terms $\mathcal{O}(\nabla\tilde{\rho}\cdot\tilde{\boldsymbol{B}})$, $\nabla\tilde{\rho}\cdot\bar{\boldsymbol{B}} = -\nabla\bar{\rho}\cdot\tilde{\boldsymbol{B}}$ yields an equation for $\tilde{\rho}$

$$\frac{\bar{\mathcal{P}}}{r}\cos\alpha\frac{\partial\tilde{\rho}}{\partial\theta} + \frac{\bar{\mathcal{I}}}{R}\frac{\partial\tilde{\rho}}{\partial\phi} = -\frac{\tilde{B}_n}{\bar{B}_n}\tilde{\mathcal{R}}\,\|\boldsymbol{\nabla}\bar{\rho}\|\,. \quad (40)$$

With the chosen coordinate system, the plasma description in the scrape-off layer, beyond the separatrix, is somewhat arbitrary. A simple linear model for the density and temperature of each species s is generally assumed

$$n_s[T_s](\rho) = \bar{n}_s\left[\bar{T}_s\right]_{(\bar{\rho}=1)} + \frac{\partial\bar{n}_s}{\partial\bar{\rho}}\left[\frac{\partial\bar{T}_s}{\partial\bar{\rho}}\right]_{(\bar{\rho}=1)}(\rho-1) \quad (41)$$

from $\rho = 1$ to $\rho = \rho^{\bar{n}_s[\bar{T}_s]}_{\max}$ with

$$\rho^{\bar{n}_s[\bar{T}_s]}_{\max} = \rho\,(n_s[T_s]=0) = -\frac{\bar{n}_s\left[\bar{T}_s\right]_{(\bar{\rho}=1)}}{\partial\bar{n}_s/\partial\bar{\rho}\left[\partial\bar{T}_s/\partial\bar{\rho}\right]_{(\bar{\rho}=1)}} + 1 \quad (42)$$

then $n_s[T_s]_{(\rho=\rho^{\bar{n}_s[\bar{T}_s]}_{\max})} = 0$ $n_s[T_s]_{(\rho=\rho^{\bar{n}_s[\bar{T}_s]}_{\max})} = 0$ for $\rho > \rho^{\bar{n}_s[\bar{T}_s]}_{\max}$, Otherwise, $n_s[T_s](\rho>1) = \bar{n}_s[\bar{T}_s]_{(\bar{\rho}=1)}$ is chosen uniformly.

Finally,

$$\frac{\partial n_s}{\partial\boldsymbol{Y}}\left[\frac{\partial T_s}{\partial\boldsymbol{Y}}\right] = \frac{\partial n_s}{\partial\rho}\left[\frac{\partial T_s}{\partial\rho}\right]\times\frac{\partial\rho}{\partial\boldsymbol{Y}} \quad (43)$$

where $\boldsymbol{Y}$ is the pseudo-vector (3) used in the ray tracing description.

The reference axisymmetric MHD equilibrium used for the calculation is described in section 6.2. The effect of magnetic ripple on LH wave propagation is calculated assuming $N = 18$ and $\bar{a}_r = 1.25$ m. With these parameters, the radial shift of nested iso-ripple surfaces is $L_0 \approx 0.256$ m and $\delta_n \approx 2.22\times10^{-3}$, such that $\delta_\rho(1,0) \approx 0.259$, $\delta_\theta(1,0) = 0$ and $\delta_\phi(1,0) \approx 0.235$. Two rays are launched at $\bar{\rho}_0 = 0.968$, slightly above the equatorial mid-plane ($\theta_0 = \pi/12$) and between two coils at $\phi_0 = \pi/24$. The initial wave characteristics at launch are given in section 6.2.

As shown in figure 5, the fast modulation on the parallel refractive index reflects the effect of the discreteness of the magnetic system. A weak modulation of the radial position of the ray is also visible on the LFS edge where the ripple amplitude is maximum. Because of the strong chaotic nature of the LH wave dynamics in a toroidal magnetic configuration, ray paths with and without magnetic ripple are completely different. However, the condition $\mathcal{D} \approx 0$ remains well satisfied regardless of amplitude of ripple effects, even when rays are reflected at the LFS edge cut-off. This important numerical aspect is already discussed in section 6.2.

7.2. Plasma fluctuations and LH wave propagation

The propagation of RF waves could be strongly modified by plasma fluctuations. This problem was addressed some time ago for the LH wave and more recently for the EC wave [21, 35–40]. In those works, the effect of fluctuations is described in terms of ray scattering by solving a wave kinetic equation in Fourier space assuming that weak fluctuations are equivalent to applying a random phase to the mode coupling [35, 41, 42]. As the wave scatters, its wave vector is rotated by

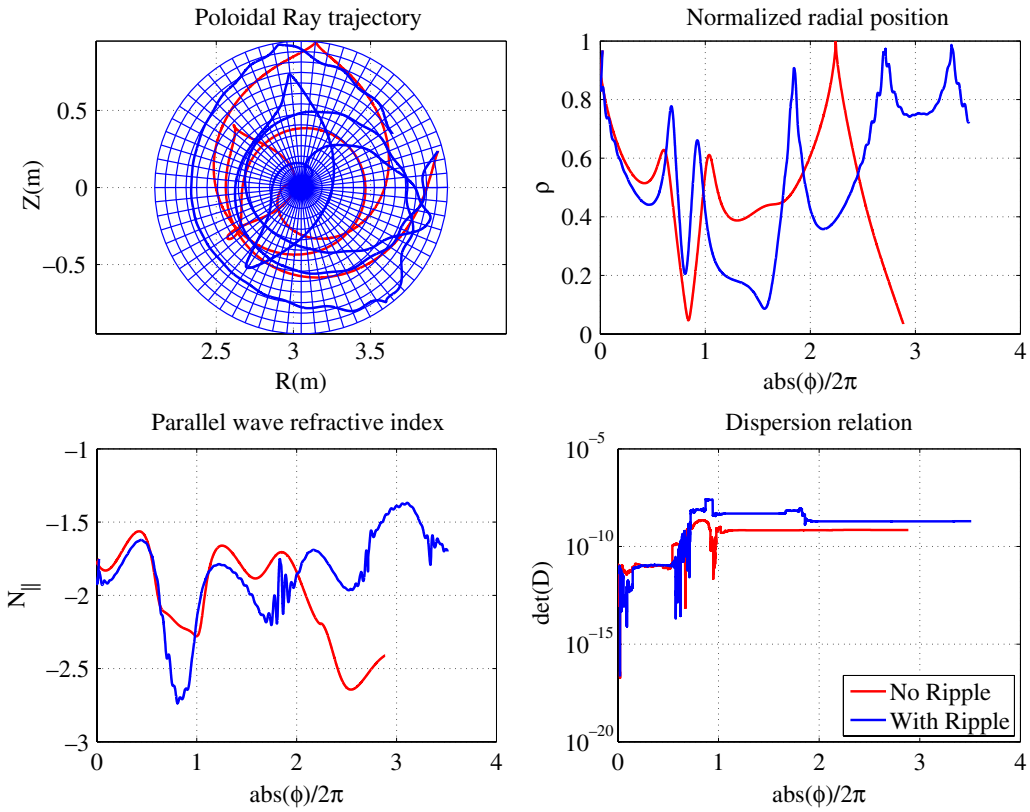


Figure 5. Poloidal projection of the ray trajectory for the LH wave calculated by $C3PO$ with $N_{\phi 0} = -1.8$ (top left). The blue and red curves correspond to the cases with and without magnetic ripple respectively. Calculations are performed with piecewise cubic Hermite interpolating polynomials for the toroidal MHD equilibrium. The effect of the discreteness of the magnetic system is clearly observed by the fast modulation on the parallel refractive index (bottom left), radial position of the ray trajectory (top right) and the dispersion relation (bottom right).

an angle β in the plane perpendicular to $\hat{\boldsymbol{b}}$. The magnitude of $\boldsymbol{k}_\perp$ remains unchanged only when the scattering occurs between like modes. However, in poor accessibility conditions for the LH wave, its amplitude may change by scattering when it occurs in a region where unlike modes can propagate [2], and $k_\perp$ acquires a new value in agreement with the dispersion relation for the mode into which the wave scatters. In this approach, any deviation from the ray trajectory calculated in the unperturbed MHD equilibrium results from a succession of discrete random scattering events, which modify the spectral properties of the rf wave [21, 38].

In $C3PO$, an alternative approach is considered, in which plasma fluctuations are simply described as a static perturbation of the toroidal MHD equilibrium [13]. Indeed, the group velocity of rf waves is always large enough such that the condition $v_g/L_D \gg \tilde{\omega}$ is well satisfied, where L_D is the ray length to complete power absorption and $\tilde{\omega}$ the fluctuation frequency spectrum of interest. The kernel of ray-tracing equations (1) and (2) remains therefore unchanged since fluctuations are described in the real space. This method has several fundamental advantages over the scattering technique, in addition to the simplicity of its numerical implementation. First, all rays launched at the same time from different positions have a consistent trajectory with respect to the spatial structure of the fluctuation modes, instead of being scattered independently. Moreover, the perturbation may be directly obtained from gyrokinetic calculations, to investigate the effect of plasma turbulence on rf wave propagation and absorption. This general method is valid as far as the WKB approximation is satisfied, i.e. $\nabla\tilde{\mathcal{A}}|_{[\rho,\theta,\phi]}/\tilde{\mathcal{A}} \ll k_{[\rho,\theta,\phi]}$, where $\tilde{\mathcal{A}}$ is the perturbation of any physical quantity $\mathcal{A}$, and the surface of constant phase remains smooth. This condition, which is nearly always verified along the ray trajectories for the EC waves, is more restrictive for the LH waves because of its longer wavelength, and sets an upper limit on the fluctuation k-spectrum that can be included in ray-tracing calculations.

In order to illustrate this method, a simple model for the electron density fluctuation is presented. The aim is to demonstrate that ray trajectories in a perturbed magnetic equilibrium can be described by ray-tracing methods while satisfying the dispersion relation. More refined models are also implemented in $C3PO$, which can describe fluctuations of the magnetic field.

For fluctuations characterized by a Gaussian 3D auto-correlation function, the perturbed electron density may be expressed in the following form:

$$\frac{\tilde{n}_e(\bar{\rho},\theta,\phi)}{\bar{n}_e(\bar{\rho})} = \tilde{F}_\lambda(\theta)\,\tilde{F}_{\mu,\Delta}(\bar{\rho}) \times \left[4\sigma_n\pi^{3/4}\sqrt{\epsilon_{\tilde{\rho}}\epsilon_\theta\epsilon_\phi} \times \sum_{\tilde{l},\tilde{m},\tilde{n}} \exp\left(-\frac{\pi^2}{2}\left(\tilde{l}^2\epsilon_{\tilde{\rho}}^2 + \tilde{m}^2\epsilon_\theta^2 + \tilde{n}^2\epsilon_\phi^2\right)\right)\sin\Phi_{\tilde{l},\tilde{m},\tilde{n}}\right] \tag{44}$$

where $(\tilde{l}, \tilde{m}, \tilde{n})$ are integers corresponding to fluctuation modes ranging between $(\tilde{l}_{\min}, \tilde{m}_{\min}, \tilde{n}_{\min})$ and $(\tilde{l}_{\max}, \tilde{m}_{\max}, \tilde{n}_{\max})$, respectively, $\Phi_{\tilde{l},\tilde{m},\tilde{n}} = 2\pi\tilde{l}\bar{\rho}+\tilde{m}\theta+\tilde{n}\phi+\varphi_{\tilde{l},\tilde{m},\tilde{n}}$ is the fluctuation phase while $\varphi_{\tilde{l},\tilde{m},\tilde{n}}$ is a uniform random number in the interval $[0, 2\pi]$. Since $\widetilde{n}_e(\bar{\rho}, \theta, \phi)$ may be larger than $\overline{n}_e(\bar{\rho})$, the quantity $1 + \widetilde{n}_e(\bar{\rho}, \theta, \phi)/\overline{n}_e(\bar{\rho})$ may become negative, and an approximate expression is considered instead, $n_e(\bar{\rho}, \theta, \phi) = \overline{n}_e(\bar{\rho})(1 + \mathcal{F}(\widetilde{n}_e(\bar{\rho}, \theta, \phi)/\overline{n}_e(\bar{\rho})))$, where $\mathcal{F}(x) = x$ when $x \geqslant 0$, and $\mathcal{F}(x) = \tanh(x)$ when $x < 0$. The function $\mathcal{F}(x)$ and its first derivatives are continuous, a requirement for ray-tracing calculations, and $\mathcal{F}(x) > -1$ everywhere. As far as $|x| \leqslant 0.5$, $\mathcal{F}(x) \approx x$ as expected for fluctuations whose amplitudes are small enough.

In (44), $\tilde{F}_\lambda(\theta) = [1 + \lambda + (1 - \lambda)\cos\theta]/2$ and $\tilde{F}_{\mu,\Delta}(\bar{\rho}) = \exp[-\ln 2(\bar{\rho} - \mu)^2/\Delta^2]$ are poloidal and radial form factors, respectively. $\tilde{F}_\lambda(\theta)$ is introduced for describing the smooth poloidal dependence of the electron density fluctuations which are found larger on the LFS (ballooning effect), μ is the radial position where the amplitude of the fluctuations is maximal, and the radial dependence is characterized by the half width at half maximum (HWHM) parameter Δ. The coefficients $(\epsilon_{\bar{\rho}}, \epsilon_\theta, \epsilon_\phi)$, which are correlation lengths normalized to reference lengths in each direction, $L_{\bar{\rho}} = a$, $L_\theta(\bar{\rho}) = \int_0^{2\pi} r/\cos\alpha \, \mathrm{d}\theta$ and $L_\phi(\bar{\rho}, \theta) = 2\pi R(\bar{\rho}, \theta)$, are chosen to be constant for the sake of simplicity. When $\tilde{F}_\lambda(\theta) = \tilde{F}_{\mu,\Delta}(\bar{\rho}) = 1$, the standard deviation of the random variable $\tilde{n}_e/\bar{n}_e$ is given by σ_n.

The calculations are performed with the reference axisymmetric toroidal MHD equilibrium presented in section 6.2 and the same LH wave parameters. Electron density fluctuations are considered to be localized on the LFS near the plasma edge as observed experimentally [13], such that $\lambda = 0$, $\mu = 1$ and $\Delta = 0.1$, while $\sigma_n = 0.7$.

From the validity of the WKB approximation, which corresponds to the condition $\widetilde{k}_{\bar{\rho},\theta,\phi} \ll k_{\rho,\theta,\phi}$ where plasma fluctuations take place, the upper bounds for $(\tilde{l}_{\max}, \tilde{m}_{\max}, \tilde{n}_{\max})$ integers are $\tilde{l}_{\max}^{\mathrm{WKB}} \approx 100$, $\tilde{m}_{\max}^{\mathrm{WKB}} \approx 500$ and $\tilde{n}_{\max}^{\mathrm{WKB}} \approx 200$ approximately, since $N_\perp \geqslant 10$ and $N_\parallel > 1$ for the LH wave along the ray. Since the sum in (44) is taken over all distinct modes until the exponential factor in (44) reaches a level $\beta \ll 1$, the condition of applicability of the Fourier series is $[\tilde{l}_{\max}, \tilde{m}_{\max}, \tilde{n}_{\max}] \approx \sqrt{-2\ln\beta}/(\pi\epsilon_{[\bar{\rho},\theta,\phi]})$. Therefore, for a given value of β, ray-tracing calculations in the presence of fluctuations may be carried out if the normalized correlation lengths in each direction fulfill the conditions $\epsilon_{[\bar{\rho},\theta,\phi]} \geqslant \sqrt{-2\ln\beta}/(\pi[\tilde{l}_{\max}^{\mathrm{WKB}}, \tilde{m}_{\max}^{\mathrm{WKB}}, \tilde{n}_{\max}^{\mathrm{WKB}}])$. Taking $\beta = 10^{-2}$ with $\epsilon_{\bar{\rho}} = \epsilon_\theta = 2 \times 10^{-2}$ and $\epsilon_\phi = 0.2$, $\tilde{l}_{\max} = \tilde{m}_{\max} = 50$ and $\tilde{n}_{\max} = 5$, which is fully consistent with the WKB approximation. Furthermore, $\tilde{l}_{\min} = \tilde{m}_{\min} = \tilde{n}_{\min} = 1$ is considered in the simulation.

In order to present reproducible numerical results, the random phase $\varphi_{\tilde{l},\tilde{m},\tilde{n}}$ is generated by the standard Mersenne twister pseudorandom number generator $\varphi_{\tilde{l},\tilde{m},\tilde{n}} = \pi(2\mathcal{M}(j_{\tilde{l},\tilde{m},\tilde{n}}) - 1)$ from its initial state with $j_{\tilde{l},\tilde{m},\tilde{n}} = \Delta\tilde{l}\Delta\tilde{m}(\tilde{n} - \tilde{n}_{\min}) + \Delta\tilde{l}(\tilde{m} - \tilde{m}_{\min}) + (\tilde{l} - \tilde{l}_{\min}) + 1$, where $\Delta[\tilde{l}, \tilde{m}, \tilde{n}] = [\tilde{l}, \tilde{m}, \tilde{n}]_{\max} - [\tilde{l}, \tilde{m}, \tilde{n}]_{\min} + 1$ are the number of modes in the Fourier series [43]. The first three coefficients of $\mathcal{M}$ in its initial state are 0.814 723 686 393 179, 0.905 791 937 075 619 and 0.126 986 816 293 506.

The ray is launched at $\bar{\rho}_0 = 0.968$, slightly above the equatorial mid-plane with $\theta_0 = \pi/12$ and $\phi_0 = 0$, and the initial spectral characteristics at launch are given in section 6.2. As shown in figure 6, the ray dynamics for the LH wave is strongly modified by the fluctuations of electron density, while the dispersion relation remains satisfied along the ray trajectory, including at the plasma cut-off. This simulation highlights the very large sensitivity of the LH wave to a perturbation of the toroidal MHD equilibrium. For the parameters considered here, the calculations are about three times longer in the presence of density fluctuations. The problem of ray propagation of EC waves in a fluctuating plasma has also been addressed using *C3PO*, for current drive calculations [13].

7.3. EC wave in shaped plasma

The accurate prediction of EC wave propagation in ITER is critical to an assessment of its ability to be used for suppressing or reducing the size of neoclassical tearing modes. In this case, the toroidal MHD equilibrium, which corresponds to the reference scenario II, is fully numeric with a strong poloidal shaping, as shown in figure 7 [44]. The ray starting location is $R = 6.4848$ m and $Z = 4.110$ m, and the vertical and toroidal starting angles are $\alpha = -63.5°$ and $\beta = 22.0°$, respectively, according to a EC benchmarking procedure given in [45]. Even if the toroidal MHD equilibrium used by *C3PO* ray tracing differs slightly from the one used in [45], results shown in figure 7 are very similar. In this case, the ray is stopped when the wave is fully linearly damped in the plasma. Here again, the dispersion relation is well satisfied along the trajectory. This example illustrates the versatility of the *C3PO* code and its applicability to the EC wave propagation [46], even though effects such as beam focusing and diffraction are not described in this pure ray-tracing scheme and require a beam-tracing formalism. It is important to note, however, that many features developed in this paper are also applicable to beam-tracing numerical schemes.

7.4. LH wave in a reverse field pinch

In RFP, the plasma evolves to a quasi-stationary toroidal MHD equilibrium close to the force free minimum energy states described by Taylor [47]. The fully relaxed Taylor states are characterized by $\nabla \times \bar{\boldsymbol{B}} = \lambda\bar{\boldsymbol{B}}$ with $\lambda = \mu\bar{\boldsymbol{J}} \cdot \bar{\boldsymbol{B}}/\bar{B}^2$ uniform throughout the plasma. Here, μ is the magnetic permittivity of the plasma. A simplified approach considers a circular concentric toroidal plasma such that $r = \bar{\rho}a$ and a constant λ value, such that

$$-\frac{1}{R}\frac{\partial}{\partial r}\left(RB_\phi\right) = \lambda B_\theta \tag{45}$$

$$\frac{1}{r}\frac{\partial}{\partial r}\left(rB_\theta\right) = \lambda B_\phi. \tag{46}$$

After eliminating B_θ, the equation in B_ϕ

$$\frac{\partial^2}{\partial r^2}\left(RB_\phi\right) + \frac{R}{r}\frac{\partial}{\partial r}\left(\frac{r}{R}\right)\frac{\partial}{\partial r}\left(RB_\phi\right) + \lambda^2 RB_\phi = 0 \tag{47}$$

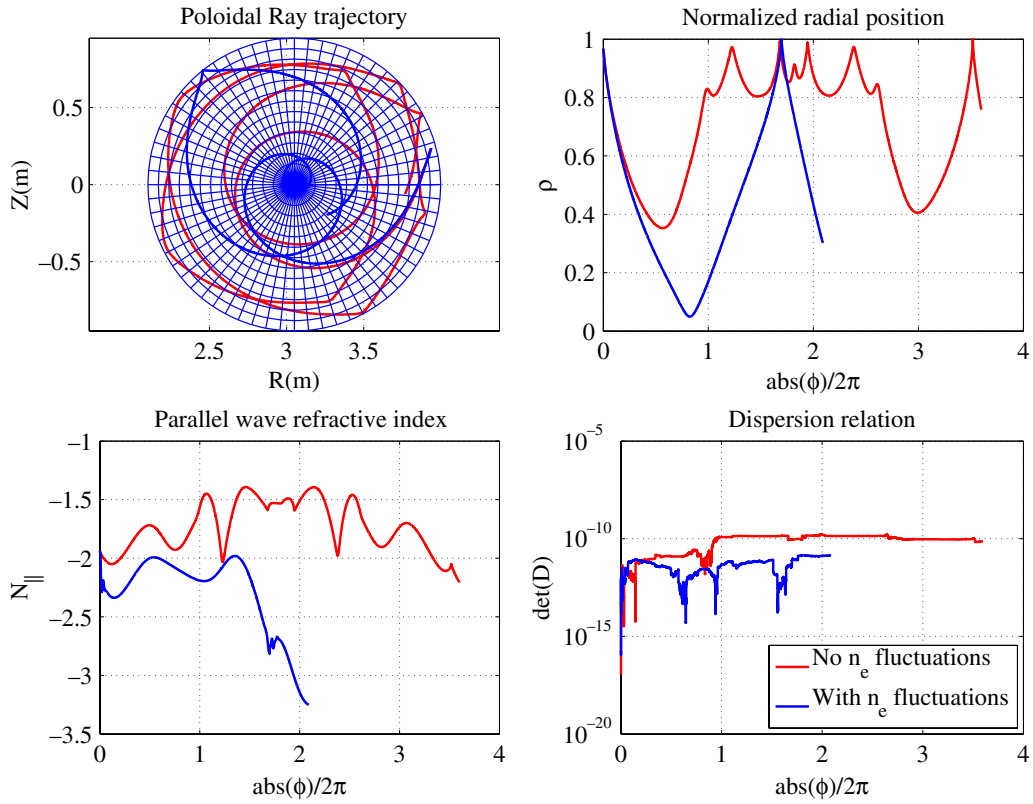


Figure 6. Poloidal projection of the ray trajectory for the LH wave calculated by $C3PO$ with $N_{\phi 0} = -1.8$ (top left). The blue and red curves correspond to the cases with and without density fluctuations, respectively. Calculations are performed with piecewise cubic Hermite interpolating polynomials for the toroidal MHD equilibrium. The effect of the density fluctuations, though localized at the LFS plasma edge, is clearly observed on the parallel refractive index which exhibits a large upshift as compared with the quiescent case (bottom left). The radial position of the ray trajectory is also strongly modified by the presence of density fluctuations (top right). However, the dispersion relation is always well satisfied, even when the ray is reflected at the edge cut-off (bottom right).

simplifies to

$$\frac{\partial^2}{\partial r^2}\left(RB_\phi\right) + \frac{1}{r}\frac{\partial}{\partial r}\left(RB_\phi\right) + \lambda^2 RB_\phi = 0 \qquad (48)$$

assuming a low inverse aspect ratio limit and neglecting all terms higher than $(a/R_0)^2$, $\partial(r/R)/\partial r \simeq 1/R$. Therefore, (47) becomes a Bessel equation, whose solution is $RB_\phi = J_0(\lambda r)$. Reporting this equation in (45) and using the derivative identity for Bessel functions of order m, one finds $RB_\theta = -J_{-1}(\lambda r) = J_{+1}(\lambda r)$ and from the determination turns out that $(\nabla \times \boldsymbol{B})_r = 0$, which means that $\nabla \times \bar{\boldsymbol{B}} = \lambda \bar{\boldsymbol{B}}$ is well satisfied. Introducing $\lambda a = \tilde{\lambda}$, up to the same order, the safety factor is

$$q \simeq \frac{a}{R_0}\rho\frac{J_0\left(\tilde{\lambda}\bar{\rho}\right)}{J_1\left(\tilde{\lambda}\bar{\rho}\right)}. \qquad (49)$$

For the ray-tracing calculations, the toroidal MHD equilibrium parameters are $R_0 = 1.5\,\mathrm{m}$, $a = 0.5\,\mathrm{m}$, while the toroidal magnetic field on axis is $\bar{B}_{T0} = 0.2\,\mathrm{T}$ and the toroidal current is $I_p = 0.3\,\mathrm{MA}$. The current density profile is derived from the MHD equilibrium with $\tilde{\lambda} = 2.2$, while the electron density profile is $n_e(\bar{\rho}) = (n_{e0} - n_{ea})(1 - \bar{\rho}^2)^{\alpha_{n_e}} + n_{ea}$ with $n_{e0} = 1.4 \times 10^{19}\,\mathrm{m}^{-3}$, $n_{ea} = 1.0 \times 10^{17}\,\mathrm{m}^{-3}$ and $\alpha_{n_e} = 1.3$. The main ion species is deuterium. Its density profile is deduced from electroneutrality and a uniform effective charge $Z_{\mathrm{eff}} = 2$, using the fully stripped single carbon impurity model. All these parameters are representative of the Madison Symmetric Torus RFP [48].

The ray is launched in the slow mode from the position $\bar{\rho}_0 = 0.99$, $\theta_0 = \pi$ and $\phi_0 = 0$ with the spectral properties $\omega/(2\pi) = 0.8\,\mathrm{GHz}$, $N_{\|0} = 10$ and $n_0 = 0$. It is worth noting that since at the plasma edge of a RFP, magnetic field lines are almost in the poloidal direction, the initial toroidal wave number n_0 is close to zero, unlike tokamaks where $m_0 \approx 0$. Because of the axisymmetry, n remains unchanged along the ray propagation even though rays may propagate toroidally. The cold plasma model is considered for the dielectric properties.

The results are shown in figure 8. A strong poloidal upshift is observed as the LH wave propagates inward, and as found by previous publications, the ray does a spiral around the magnetic axis [49]. The dispersion relation is well verified along the ray path.

8. Conclusion

Within the conditions of applicability of the WKB approximation, ray-tracing calculations are effective for describing rf wave propagation in toroidal magnetic plasmas.

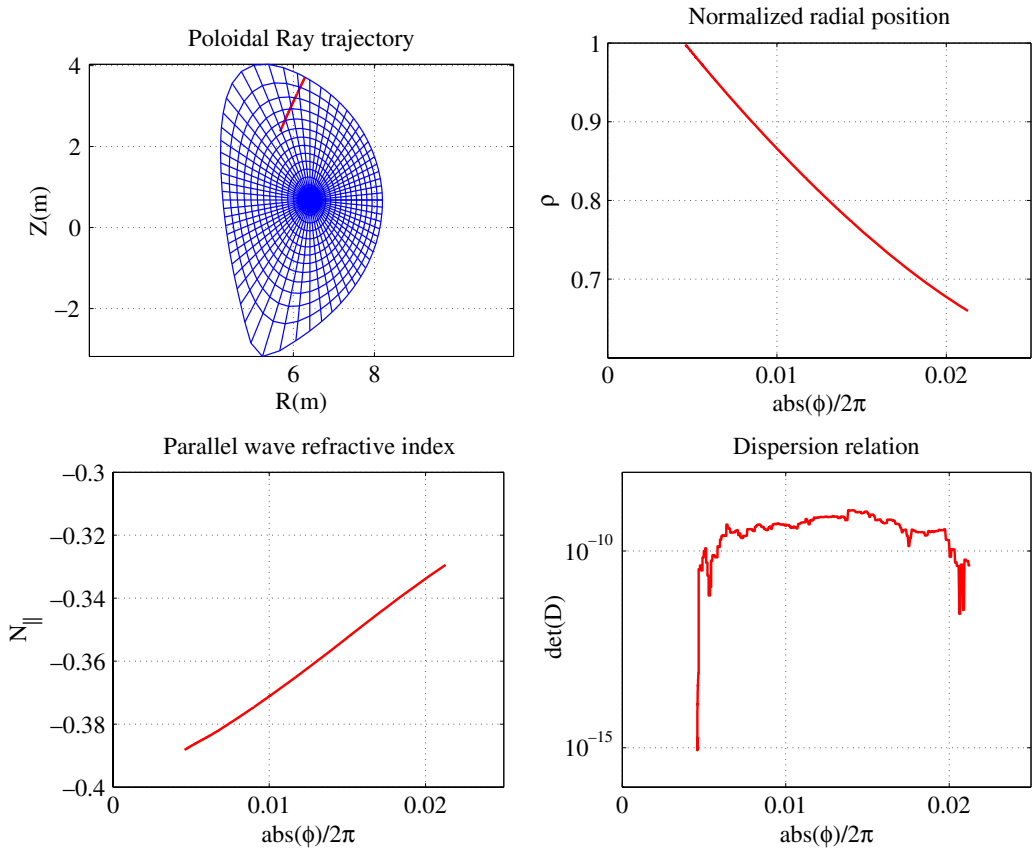


Figure 7. Poloidal projection of the ray trajectory for the EC wave calculated by $C3PO$ (top left). Calculations are performed with piecewise cubic Hermite interpolating polynomials for the ITER tokamak configuration [44]. The parallel refractive index exhibits a progressive decrease as the ray propagates inward from the top of the plasma (bottom left and top right). The dispersion relation is well satisfied along the ray trajectory (bottom right). Calculations are automatically stopped when the rf power carried by the ray is linearly absorbed by the plasma. All the results compare well with those given in [45].

In this context, a new ray-tracing code named $C3PO$ has been developed, primarily intended to be used with Fokker–Planck solvers in quasi-linear calculations of the power absorbed or the current driven in the plasma resulting from resonant wave–particle interaction.

As part of a chain of code, its design is highly optimized to obtain accurate results while reducing the computational effort. Based on the magnetic flux coordinate system which is particularly appropriate for calculations in toroidal magnetic configurations, ray trajectories are accurately determined by a Runge–Kutta–Fehlberg method for integrating the set of ordinary differential equations. As the dispersion relation is expressed in terms of a set of non-dimensional quantities, the dielectric properties of the plasma can be separated from the metrics, thus allowing to build a code with a modular structure, making it very flexible for studying rf propagation at various frequencies, in different magnetic configurations and with distinct plasma dielectric models.

Furthermore, thanks to its high accuracy and numerical efficiency in calculating interpolated values and their derivatives, it is possible to address advanced physical problems, like the effect of a discrete toroidal coil system, or that of plasma fluctuations in a fully original quasi-optical approach, on the ray trajectories, provided magnetic flux surfaces remain nested.

The code has been highly benchmarked and validated for simple examples against the *FRTC* and *DELPHINE* ray-tracing codes, while a large set of relevant examples has been given to illustrate the capabilities of this tool. The improved accuracy and performance of analytical derivatives derived from a spline-Fourier interpolation over the numerical 2D derivatives is clearly demonstrated.

Its modular structure allows various future developments and applications, such as new plasma dielectric models for studying relativistic plasma effects on the ray propagation, or other coordinate systems for different plasma configurations. Furthermore, in the physical and numerical framework of this model, additional effects may be implemented to extend the code to a beam-tracing formalism for a more realistic description of EC wave propagation.

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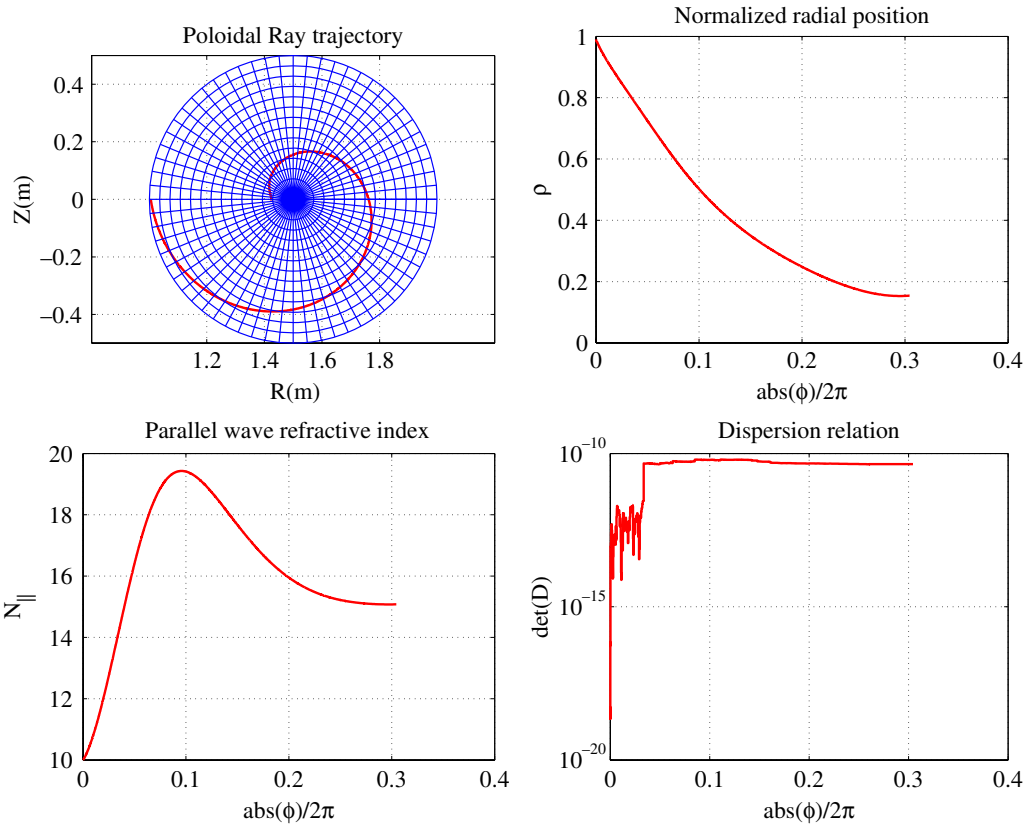


Figure 8. Poloidal projection of the ray trajectory for the LH wave calculated by $C3PO$ with $N_{\|0} = 10$ (top left). Calculations are performed with piecewise cubic Hermite interpolating polynomials for the RFP magnetic configuration. The parallel refractive index exhibits a fast increase as the ray propagates inward from the HFS (bottom left and top right). The dispersion relation is well satisfied along the ray trajectory (bottom right).

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